1) a)

$2\sinh x \equiv 2\left(\frac{e^x - e^{-x}}{2}\right)$

$\equiv e^x - e^{-x}$

1) b)

$8\cosh x \equiv 8\left(\frac{e^x + e^{-x}}{2}\right)$

$\equiv 4e^x + 4e^{-x}$

1) c)

$4\sinh x - 4\cosh x \equiv 4\left(\frac{e^x - e^{-x}}{2}\right) - 4\left(\frac{e^x + e^{-x}}{2}\right)$

$\equiv 2(e^x - e^{-x}) - 2(e^x + e^{-x})$

$\equiv 2e^x - 2e^{-x} - 2e^x - 2e^{-x}$

$\equiv -4e^{-x}$

1) d)

$5\cosh x - 10\sinh x \equiv 5\left(\frac{e^x + e^{-x}}{2}\right) - 10\left(\frac{e^x - e^{-x}}{2}\right)$

$\equiv \frac{5}{2}(e^x + e^{-x}) - 5(e^x - e^{-x})$

$\equiv \frac{5}{2}e^x + \frac{5}{2}e^{-x} - 5e^x + 5e^{-x}$

$\equiv -\frac{5}{2}e^x + \frac{15}{2}e^{-x}$

2) a)

$2\sinh x + 4\cosh x \equiv 2\left(\frac{e^x - e^{-x}}{2}\right) + 4\left(\frac{e^x + e^{-x}}{2}\right)$

$\equiv (e^x - e^{-x}) + 2(e^x + e^{-x})$

$\equiv e^x - e^{-x} + 2e^x + 2e^{-x}$

$\equiv 3e^x + e^{-x}$

$\equiv e^{-x}(3e^{2x} + 1)$

2) b)

$3\sinh x - 9\cosh x \equiv 3\left(\frac{e^x - e^{-x}}{2}\right) - 9\left(\frac{e^x + e^{-x}}{2}\right)$

$\equiv \frac{3}{2}(e^x - e^{-x}) - \frac{9}{2}(e^x + e^{-x})$

$\equiv \frac{3}{2}e^x - \frac{3}{2}e^{-x} - \frac{9}{2}e^x - \frac{9}{2}e^{-x}$

$\equiv -3e^x - 6e^{-x}$

$\equiv -3e^{-x}(e^{2x} + 2)$

2) c)

$10\cosh x - 8\sinh x \equiv 10\left(\frac{e^x + e^{-x}}{2}\right) - 8\left(\frac{e^x - e^{-x}}{2}\right)$

$\equiv 5(e^x + e^{-x}) - 4(e^x - e^{-x})$

$\equiv 5e^x + 5e^{-x} - 4e^x + 4e^{-x}$

$\equiv e^x + 9e^{-x}$

$\equiv e^{-x}(e^{2x} + 9)$

2) d)

$26\cosh x + 42\sinh x \equiv 26\left(\frac{e^x + e^{-x}}{2}\right) + 42\left(\frac{e^x - e^{-x}}{2}\right)$

$\equiv 13(e^x + e^{-x}) + 21(e^x - e^{-x})$

$\equiv 13e^x + 13e^{-x} + 21e^x - 21e^{-x}$

$\equiv 34e^x - 8e^{-x}$

$\equiv 2e^{-x}(17e^{2x} - 4)$

3)

$55\sinh x - 35\cosh x \equiv 55\left(\frac{e^x - e^{-x}}{2}\right) - 35\left(\frac{e^x + e^{-x}}{2}\right)$

$\equiv \frac{55}{2}(e^x - e^{-x}) - \frac{35}{2}(e^x + e^{-x})$

$\equiv \frac{55}{2}e^x - \frac{55}{2}e^{-x} - \frac{35}{2}e^x - \frac{35}{2}e^{-x}$

$\equiv 10e^x - 45e^{-x}$

$\equiv 5e^{-x}(2e^{2x} - 9)$

4) a)

$\sinh 0 \equiv \frac{e^0 - e^{-0}}{2} \equiv \frac{1-1}{2} = 0$

4) b)

$\cosh 0 = \frac{e^{0}+e^{-0}}{2} = \frac{1+1}{2} = 1$

5) a)

$\sinh 1 = \frac{e^{1}-e^{-1}}{2} = \frac{e-\frac{1}{e}}{2} = 1.18$ (3 SF)

5) b)

$\cosh 1 = \frac{e^{1}+e^{-1}}{2} = \frac{e+\frac{1}{e}}{2} = 1.54$ (3 SF)

5) c)

$\sinh(-1) = \frac{e^{(-1)}-e^{-(-1)}}{2} = \frac{e^{-1}-e}{2} = \frac{\frac{1}{e}-e}{2} = -1.18$ (3 SF)

5) d)

$\cosh(-1) = \frac{e^{(-1)}+e^{-(-1)}}{2} = \frac{e^{-1}+e}{2} = \frac{\frac{1}{e}+e}{2} = 1.54$ (3 SF)

6)

LHS: $\sinh(-x) \equiv \frac{e^{(-x)}-e^{-(-x)}}{2} \equiv \frac{e^{-x}-e^{x}}{2} \equiv -\frac{e^{x}-e^{-x}}{2} \equiv -\sinh x$: RHS

7)

LHS: $\cosh(-x) \equiv \frac{e^{(-x)}+e^{-(-x)}}{2} \equiv \frac{e^{-x}+e^{x}}{2} \equiv \frac{e^{x}+e^{-x}}{2} \equiv \cosh x$: RHS

8) a)

$\sinh(\ln 2) = \frac{e^{\ln 2}-e^{-\ln 2}}{2} = \frac{e^{\ln 2}-e^{\ln 2^{-1}}}{2} = \frac{2-2^{-1}}{2} = \frac{3}{4}$

8) b)

$\cosh(\ln 3) = \frac{e^{\ln 3}+e^{-\ln 3}}{2} = \frac{e^{\ln 3}+e^{\ln 3^{-1}}}{2} = \frac{3+3^{-1}}{2} = \frac{5}{3}$

8) c)

$$\sinh\left(\ln\frac{1}{3}\right)=\frac{e^{\ln\frac{1}{3}}-e^{-\ln\frac{1}{3}}}{2}=\frac{e^{\ln\frac{1}{3}}-e^{\ln\left(\frac{1}{3}\right)^{-1}}}{2}=\frac{\frac{1}{3}-\left(\frac{1}{3}\right)^{-1}}{2}=\frac{\frac{1}{3}-3}{2}=-\frac{4}{3}$$

8) d)

$$\cosh\left(\ln\frac{1}{4}\right)=\frac{e^{\ln\frac{1}{4}}+e^{-\ln\frac{1}{4}}}{2}=\frac{e^{\ln\frac{1}{4}}+e^{\ln\left(\frac{1}{4}\right)^{-1}}}{2}=\frac{\frac{1}{4}+\left(\frac{1}{4}\right)^{-1}}{2}=\frac{\frac{1}{4}+4}{2}=\frac{17}{8}$$

9)

$$\sinh(\ln k)=\frac{e^{\ln k}-e^{-\ln k}}{2}=\frac{e^{\ln k}-e^{\ln k^{-1}}}{2}=\frac{k-\frac{1}{k}}{2}=\frac{k^2-1}{2k}$$

10)

$$\cosh(2\ln k)=\frac{e^{2\ln k}+e^{-2\ln k}}{2}=\frac{e^{\ln k^2}+e^{\ln k^{-2}}}{2}=\frac{k^2+\frac{1}{k^2}}{2}=\frac{k^4+1}{2k^2}$$

11)

$$\tanh x\equiv\frac{\sinh x}{\cosh x}\equiv\frac{\frac{e^x-e^{-x}}{2}}{\frac{e^x+e^{-x}}{2}}\equiv\frac{e^x-e^{-x}}{e^x+e^{-x}}\equiv\frac{e^{2x}-1}{e^{2x}+1}\equiv\frac{(e^x+1)(e^x-1)}{e^{2x}+1}$$

12)

$$\text{LHS:}\frac{1}{2}\ln\left(\frac{1+\tanh x}{1-\tanh x}\right)=\frac{1}{2}\ln\left(\frac{1+\frac{\sinh x}{\cosh x}}{1-\frac{\sinh x}{\cosh x}}\right)=\frac{1}{2}\ln\left(\frac{\cosh x+\sinh x}{\cosh x-\sinh x}\right)$$

$$=\frac{1}{2}\ln\left(\frac{\frac{e^x+e^{-x}}{2}+\frac{e^x-e^{-x}}{2}}{\frac{e^x+e^{-x}}{2}-\frac{e^x-e^{-x}}{2}}\right)=\frac{1}{2}\ln\left(\frac{e^x+e^{-x}+e^x-e^{-x}}{e^x+e^{-x}-(e^x-e^{-x})}\right)$$

$$=\frac{1}{2}\ln\left(\frac{2e^x}{2e^{-x}}\right)=\frac{1}{2}\ln\left(\frac{e^x}{e^{-x}}\right)=\frac{1}{2}\ln\left(e^{x-(-x)}\right)=\frac{1}{2}\ln(e^{2x})=\frac{1}{2}\times 2x=x:\text{RHS}$$

13) a)

$$\tanh 0=\frac{e^{2\times 0}-1}{e^{2\times 0}+1}=\frac{1-1}{1+1}=0$$

13) b)

$$\tanh 1=\frac{e^{2\times 1}-1}{e^{2\times 1}+1}=\frac{e^2-1}{e^2+1}$$

13) c)

$$\tanh(-1) = \frac{e^{2\times(-1)}-1}{e^{2\times(-1)}+1} = \frac{e^{-2}-1}{e^{-2}+1} = \frac{1-e^2}{1+e^2}$$

14)

$$\tanh(-x) \equiv \frac{e^{2(-x)}-1}{e^{2(-x)}+1} \equiv \frac{e^{-2x}-1}{e^{-2x}+1} \equiv \frac{1-e^{2x}}{1+e^{2x}} \equiv -\frac{e^{2x}-1}{e^{2x}+1} = -\tanh x$$

Alternative Method:

$$\tanh(-x) \equiv \frac{\sinh(-x)}{\cosh(-x)} \equiv \frac{-\sinh x}{\cosh x} \equiv -\tanh x$$

15) a)

$$\tanh(\ln 3) = \frac{e^{2(\ln 3)}-1}{e^{2(\ln 3)}+1} = \frac{e^{\ln 3^2}-1}{e^{\ln 3^2}+1} = \frac{3^2-1}{3^2+1} = \frac{4}{5}$$

15) b)

$$\tanh(-\ln 4) = \frac{e^{2(-\ln 4)}-1}{e^{2(-\ln 4)}+1} = \frac{e^{(-2\ln 4)}-1}{e^{(-2\ln 4)}+1} = \frac{e^{\ln 4^{-2}}-1}{e^{\ln 4^{-2}}+1} = \frac{4^{-2}-1}{4^{-2}+1} = -\frac{15}{17}$$

15) c)

$$\tanh\left(\ln\frac{1}{5}\right) = \frac{e^{2\ln\frac{1}{5}}-1}{e^{2\ln\frac{1}{5}}+1} = \frac{e^{\ln\left(\frac{1}{5}\right)^2}-1}{e^{\ln\left(\frac{1}{5}\right)^2}+1} = \frac{\left(\frac{1}{5}\right)^2-1}{\left(\frac{1}{5}\right)^2+1} = -\frac{12}{13}$$

16)

$$\sinh x + \cosh x \equiv \frac{e^x-e^{-x}}{2} + \frac{e^x+e^{-x}}{2}$$

$$\equiv \frac{1}{2}(e^x - e^{-x} + e^x + e^{-x}) \equiv \frac{1}{2}(2e^x) \equiv e^x > 0 \text{ for all real values of } x$$

17)

$$\sinh x - \cosh x \equiv \frac{e^x-e^{-x}}{2} - \frac{e^x+e^{-x}}{2}$$

$$\equiv \frac{1}{2}(e^x - e^{-x} - e^x - e^{-x}) \equiv \frac{1}{2}(-2e^{-x}) \equiv -e^{-x} < 0 \text{ for all real values of } x \text{ as } e^{-x} > 0$$

18) a)

$$\sinh x = 0 \Rightarrow \frac{e^x-e^{-x}}{2} = 0 \Rightarrow e^x - e^{-x} = 0$$

$$\Rightarrow e^{2x} - 1 = 0 \Rightarrow e^{2x} = 1 \Rightarrow 2x = 0 \Rightarrow x = 0$$

18) b)

$\sinh x = 1 \Rightarrow \frac{e^x - e^{-x}}{2} = 1$

$\Rightarrow e^x - e^{-x} = 2$

$\Rightarrow e^{2x} - 1 = 2e^x$

$\Rightarrow e^{2x} - 2e^x - 1 = 0$

$\Rightarrow (e^x - 1)^2 - 1 - 1 = 0$

$\Rightarrow (e^x - 1)^2 = 2$

$\Rightarrow e^x = 1 \pm \sqrt{2}$

$\Rightarrow x = \ln(1 + \sqrt{2})$ or $x = \ln(1 - \sqrt{2})$

However, $1 - \sqrt{2} < 0$, so the only real solution is $x = \ln(1 + \sqrt{2})$

18) c)

$\sinh x = -10 \Rightarrow \frac{e^x - e^{-x}}{2} = -10$

$\Rightarrow e^x - e^{-x} = -20$

$\Rightarrow e^{2x} - 1 = -20e^x$

$\Rightarrow e^{2x} + 20e^x - 1 = 0$

$\Rightarrow (e^x + 10)^2 - 10^2 - 1 = 0$

$\Rightarrow (e^x + 10)^2 = 101$

$\Rightarrow e^x = -10 \pm \sqrt{101}$

$\Rightarrow x = \ln(-10 + \sqrt{101})$ or $x = \ln(-10 - \sqrt{101})$

However, $-10 - \sqrt{101} < 0$, so the only real solution is $x = \ln(-10 + \sqrt{101})$

19) a)

$\cosh x = 1 \Rightarrow \frac{e^x + e^{-x}}{2} = 1$

$\Rightarrow e^x + e^{-x} = 2$

$\Rightarrow e^{2x} + 1 = 2e^x$

$\Rightarrow e^{2x} - 2e^x + 1 = 0$

$\Rightarrow (e^x - 1)^2 - 1^2 + 1 = 0$

$\Rightarrow (e^x - 1)^2 = 0$

$\Rightarrow e^x = 1$

$\Rightarrow x = 0$

19) b)

$\cosh x = 5 \Rightarrow \frac{e^x + e^{-x}}{2} = 5$

$\Rightarrow e^x + e^{-x} = 10$

$\Rightarrow e^{2x} + 1 = 10e^x$

$\Rightarrow e^{2x} - 10e^x + 1 = 0$

$\Rightarrow (e^x - 5)^2 - 5^2 + 1 = 0$

$\Rightarrow (e^x - 5)^2 = 24$

$\Rightarrow e^x = 5 \pm 2\sqrt{6}$

$\Rightarrow x = \ln(5 + 2\sqrt{6})$ or $x = \ln(5 - 2\sqrt{6})$

19) c)

$\cosh x = 23 \Rightarrow \frac{e^x + e^{-x}}{2} = 23$

$\Rightarrow e^x + e^{-x} = 46$

$\Rightarrow e^{2x} + 1 = 46e^x$

$\Rightarrow e^{2x} - 46e^x + 1 = 0$

$\Rightarrow (e^x - 23)^2 - 23^2 + 1 = 0$

$\Rightarrow (e^x - 23)^2 = 528$

$\Rightarrow e^x = 23 \pm 4\sqrt{33}$

$\Rightarrow x = \ln(23 + 4\sqrt{33})$ or $x = \ln(23 - 4\sqrt{33})$

20) a)

$\tanh x = 0 \Rightarrow \frac{e^{2x} - 1}{e^{2x} + 1} = 0$

$\Rightarrow e^{2x} - 1 = 0$

$\Rightarrow e^{2x} = 1$

$\Rightarrow 2x = 0$

$\Rightarrow x = 0$

20) b)

$\tanh x = \frac{1}{3} \Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = \frac{1}{3}$

$\Rightarrow 3e^{2x} - 3 = e^{2x} + 1$

$\Rightarrow 2e^{2x} = 4$

$\Rightarrow e^{2x} = 2$

$\Rightarrow 2x = \ln 2$

$\Rightarrow x = \frac{1}{2}\ln 2$

20) c)

$\tanh x = -\frac{4}{5} \Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = -\frac{4}{5}$

$\Rightarrow 5e^{2x} - 5 = -4e^{2x} - 4$

$\Rightarrow 9e^{2x} = 1$

$\Rightarrow e^{2x} = \frac{1}{9}$

$\Rightarrow 2x = \ln\frac{1}{9}$

$\Rightarrow x = \frac{1}{2}\ln\frac{1}{9}$

$\Rightarrow x = \ln\left(\frac{1}{9}\right)^{\frac{1}{2}} = \ln\frac{1}{3} = -\ln 3$

21) a)

$\sinh x + \cosh x = 2 \Rightarrow \frac{e^{x}-e^{-x}}{2} + \frac{e^{x}+e^{-x}}{2} = 2$

$\Rightarrow e^{x} - e^{-x} + e^{x} + e^{-x} = 4$

$\Rightarrow 2e^{x} = 4$

$\Rightarrow e^{x} = 2$

$\Rightarrow x = \ln 2$

21) b)

$2\sinh x - \cosh x = 1 \Rightarrow 2 \times \frac{e^x - e^{-x}}{2} - \frac{e^x + e^{-x}}{2} = 1$

$\Rightarrow e^x - e^{-x} - \frac{e^x + e^{-x}}{2} = 1$

$\Rightarrow 2e^x - 2e^{-x} - e^x - e^{-x} = 2$

$\Rightarrow e^x - 3e^{-x} = 2$

$\Rightarrow e^{2x} - 3 = 2e^x$

$\Rightarrow e^{2x} - 2e^x - 3 = 0$

$\Rightarrow (e^x + 1)(e^x - 3) = 0$

$\Rightarrow e^x = -1$ or $e^x = 3$

$\Rightarrow x = \ln 3$ only as $e^x \neq -1$

21) c)

$\sinh x - 2\cosh x = -2 \Rightarrow \frac{e^x - e^{-x}}{2} - 2 \times \frac{e^x + e^{-x}}{2} = -2$

$\Rightarrow \frac{e^x - e^{-x}}{2} - (e^x + e^{-x}) = -2$

$\Rightarrow e^x - e^{-x} - 2e^x - 2e^{-x} = -4$

$\Rightarrow -e^x - 3e^{-x} = -4$

$\Rightarrow -e^{2x} - 3 = -4e^x$

$\Rightarrow e^{2x} - 4e^x + 3 = 0$

$\Rightarrow (e^x - 3)(e^x - 1) = 0$

$\Rightarrow e^x = 3$ or $e^x = 1$

$\Rightarrow x = \ln 3$ or $x = 0$

22) a)

$2\cosh x - 3\sinh x = \frac{1}{2} \Rightarrow 2 \times \frac{e^x + e^{-x}}{2} - 3 \times \frac{e^x - e^{-x}}{2} = \frac{1}{2}$

$\Rightarrow e^x + e^{-x} - \frac{3}{2}(e^x - e^{-x}) = \frac{1}{2}$

$\Rightarrow 2e^x + 2e^{-x} - 3e^x + 3e^{-x} = 1$

$\Rightarrow -e^x + 5e^{-x} = 1$

$\Rightarrow -e^{2x} + 5 = e^x$

$\Rightarrow e^{2x} + e^x - 5 = 0$

$\Rightarrow \left(e^x + \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 - 5 = 0$

$\Rightarrow \left(e^x + \frac{1}{2}\right)^2 = \frac{21}{4}$

$\Rightarrow e^x = -\frac{1}{2} \pm \frac{\sqrt{21}}{2}$

$\Rightarrow x = \ln\left(-\frac{1}{2} + \frac{\sqrt{21}}{2}\right)$ or $x = \ln\left(-\frac{1}{2} - \frac{\sqrt{21}}{2}\right)$

However, $-\frac{1}{2} - \frac{\sqrt{21}}{2} < 0$, so the only real solution is $x = \ln\left(-\frac{1}{2} + \frac{\sqrt{21}}{2}\right)$

22) b)

$5\sinh x - 6\cosh x = -7 \Rightarrow 5 \times \frac{e^x - e^{-x}}{2} - 6 \times \frac{e^x + e^{-x}}{2} = -7$

$\Rightarrow 5(e^x - e^{-x}) - 6(e^x + e^{-x}) = -14$

$\Rightarrow 5e^x - 5e^{-x} - 6e^x - 6e^{-x} = -14$

$\Rightarrow -e^x - 11e^{-x} = -14$

$\Rightarrow -e^{2x} - 11 = -14e^x$

$\Rightarrow e^{2x} - 14e^x + 11 = 0$

$\Rightarrow (e^x - 7)^2 - 7^2 + 11 = 0$

$\Rightarrow (e^x - 7)^2 = 38$

$\Rightarrow e^x = 7 \pm \sqrt{38}$

$\Rightarrow x = \ln(7 + \sqrt{38})$ or $x = \ln(7 - \sqrt{38})$

22) c)

$3\cosh x - 5\sinh x = 2 \Rightarrow 3 \times \frac{e^x + e^{-x}}{2} - 5 \times \frac{e^x - e^{-x}}{2} = 2$

$\Rightarrow 3(e^x + e^{-x}) - 5(e^x - e^{-x}) = 4$

$\Rightarrow 3e^x + 3e^{-x} - 5e^x + 5e^{-x} = 4$

$\Rightarrow -2e^x + 8e^{-x} = 4$

$\Rightarrow -e^x + 4e^{-x} = 2$

$\Rightarrow -e^{2x} + 4 = 2e^x$

$\Rightarrow e^{2x} + 2e^x - 4 = 0$

$\Rightarrow (e^x + 1)^2 - 1^2 - 4 = 0$

$\Rightarrow (e^x + 1)^2 = 5$

$\Rightarrow e^x = -1 \pm \sqrt{5}$

$\Rightarrow x = \ln(-1 + \sqrt{5})$ or $x = \ln(-1 - \sqrt{5})$

However, $-1 - \sqrt{5} < 0$, so the only real solution is $x = \ln(-1 + \sqrt{5})$

22) d)

$4\sinh x + 3\cosh x = 4 \Rightarrow 4 \times \frac{e^x - e^{-x}}{2} + 3 \times \frac{e^x + e^{-x}}{2} = 4$

$\Rightarrow 2e^x - 2e^{-x} + \frac{3}{2}(e^x + e^{-x}) = 4$

$\Rightarrow 4e^x - 4e^{-x} + 3e^x + 3e^{-x} = 8$

$\Rightarrow 7e^x - e^{-x} = 8$

$\Rightarrow 7e^{2x} - 1 = 8e^x$

$\Rightarrow 7e^{2x} - 8e^x - 1 = 0$

$\Rightarrow e^{2x} - \frac{8}{7}e^x - \frac{1}{7} = 0$

$\Rightarrow \left(e^x - \frac{4}{7}\right)^2 - \left(\frac{4}{7}\right)^2 - \frac{1}{7} = 0$

$\Rightarrow \left(e^x - \frac{4}{7}\right)^2 = \frac{23}{49}$

$\Rightarrow e^x = \frac{4}{7} \pm \frac{\sqrt{23}}{7}$

$\Rightarrow x = \ln\left(\frac{4}{7} + \frac{\sqrt{23}}{7}\right)$ or $x = \ln\left(\frac{4}{7} - \frac{\sqrt{23}}{7}\right)$

However, $\frac{4}{7} - \frac{\sqrt{23}}{7} < 0$, so the only real solution is $x = \ln\left(\frac{4}{7} + \frac{\sqrt{23}}{7}\right)$

23) a)

$2\sinh 2x = 2 - 7e^{2x}$

$\Rightarrow 2 \times \frac{e^{2x} - e^{-2x}}{2} = 2 - 7e^{2x}$

$\Rightarrow e^{2x} - e^{-2x} = 2 - 7e^{2x}$

$\Rightarrow e^{4x} - 1 = 2e^{2x} - 7e^{4x}$

$\Rightarrow 8e^{4x} - 2e^{2x} - 1 = 0$

$\Rightarrow (4e^{2x} + 1)(2e^{2x} - 1) = 0$

$\Rightarrow 4e^{2x} + 1 = 0$ or $2e^{2x} - 1 = 0$

First consider $4e^{2x} + 1 = 0$:

As $e^{2x} > 0$, this equation has no real solutions.

Now consider $2e^{2x} - 1 = 0$:

$\Rightarrow e^{2x} = \frac{1}{2} \Rightarrow 2x = \ln\frac{1}{2} \Rightarrow x = \frac{1}{2}\ln\frac{1}{2}$

23) b)

$3\sinh 2x = 2e^{2x} - 2$

$\Rightarrow 3 \times \frac{e^{2x} - e^{-2x}}{2} = 2e^{2x} - 2$

$\Rightarrow 3(e^{2x} - e^{-2x}) = 4e^{2x} - 4$

$\Rightarrow 3e^{2x} - 3e^{-2x} = 4e^{2x} - 4$

$\Rightarrow 3e^{4x} - 3 = 4e^{4x} - 4e^{2x}$

$\Rightarrow e^{4x} - 4e^{2x} + 3 = 0$

$\Rightarrow (e^{2x} - 3)(e^{2x} - 1) = 0$

$\Rightarrow e^{2x} - 3 = 0$ or $e^{2x} - 1 = 0$

First consider $e^{2x} - 3 = 0$:

$\Rightarrow e^{2x} = 3 \Rightarrow 2x = \ln 3 \Rightarrow x = \frac{1}{2}\ln 3$

Now consider $e^{2x} - 1 = 0$:

$\Rightarrow e^{2x} = 1 \Rightarrow x = 0$

$\therefore x = \frac{1}{2}\ln 3$ or $x = 0$

24) a)

$(y^2 - 6y + 1)(y^2 + y + 1) \equiv y^4 + y^3 + y^2 - 6y^3 - 6y^2 - 6y + y^2 + y + 1$

$\equiv y^4 - 5y^3 - 4y^2 - 5y + 1$

24) b)

$\cosh 2x = 2 + 5\cosh x \Rightarrow \frac{e^{2x}+e^{-2x}}{2} = 2 + 5 \times \frac{e^{x}+e^{-x}}{2}$

$\Rightarrow e^{2x} + e^{-2x} = 4 + 5(e^{x} + e^{-x})$

$\Rightarrow e^{2x} + e^{-2x} = 4 + 5e^{x} + 5e^{-x}$

$\Rightarrow e^{4x} + 1 = 4e^{2x} + 5e^{3x} + 5e^{x}$

$\Rightarrow e^{4x} - 5e^{3x} - 4e^{2x} - 5e^{x} + 1 = 0$

$\Rightarrow (e^{2x} - 6e^{x} + 1)(e^{2x} + e^{x} + 1) = 0$

$\Rightarrow e^{2x} - 6e^{x} + 1 = 0$ or $e^{2x} + e^{x} + 1 = 0$

First consider $e^{2x} + e^{x} + 1 = 0$:

As $e^{2x} > 0$ and $e^{x} > 0$, this equation has no real solutions.

Now consider $e^{2x} - 6e^{x} + 1 = 0$:

$\Rightarrow (e^{x} - 3)^2 - 3^2 + 1 = 0$

$\Rightarrow (e^{x} - 3)^2 = 8$

$\Rightarrow e^{x} = 3 \pm 2\sqrt{2}$

$\Rightarrow x = \ln(3 + 2\sqrt{2})$ or $x = \ln(3 - 2\sqrt{2})$

25) a) i)

$f(1) = 1^4 - 4 \times 1^3 + 4 \times 1 - 1 = 1 - 4 + 4 - 1 = 0$

$f(-1) = (-1)^4 - 4 \times (-1)^3 + 4 \times (-1) - 1 = 1 + 4 - 4 - 1 = 0$

$\therefore$ both $(y - 1)$ and $(y + 1)$ are factors of $f(y)$ by the Factor Theorem.

25) a) ii)

$f(y) = y^4 - 4y^3 + 4y - 1$

$\equiv (y - 1)(y^3 - 3y^2 - 3y + 1)$

$\equiv (y - 1)(y + 1)(y^2 - 4y + 1)$

25) b)

$\sinh 2x = 4\sinh x \Rightarrow \frac{e^{2x}-e^{-2x}}{2} = 4 \times \frac{e^{x}-e^{-x}}{2}$

$\Rightarrow e^{2x} - e^{-2x} = 4(e^{x} - e^{-x})$

$\Rightarrow e^{2x} - e^{-2x} = 4e^{x} - 4e^{-x}$

$\Rightarrow e^{4x} - 1 = 4e^{3x} - 4e^{x}$

$\Rightarrow e^{4x} - 4e^{3x} + 4e^{x} - 1 = 0$

$\Rightarrow (e^{x} - 1)(e^{x} + 1)(e^{2x} - 4e^{x} + 1) = 0$

$\Rightarrow e^{x} - 1 = 0$ or $e^{x} + 1 = 0$ or $e^{2x} - 4e^{x} + 1 = 0$

First consider $e^{x} + 1 = 0$:

As $e^{x} > 0$, this equation has no real solutions.

Now consider $e^{x} - 1 = 0$:

$\Rightarrow e^{x} = 1 \Rightarrow x = 0$

Now consider $e^{2x} - 4e^{x} + 1 = 0$:

$\Rightarrow (e^{x} - 2)^2 - 2^2 + 1 = 0$

$\Rightarrow (e^{x} - 2)^2 = 3$

$\Rightarrow e^{x} = 2 \pm \sqrt{3}$

$\Rightarrow x = \ln(2 + \sqrt{3})$ or $x = \ln(2 - \sqrt{3})$

So the solutions to $\sinh 2x = 4\sinh x$ are $x = 0$, or $x = \ln(2 + \sqrt{3})$ or $x = \ln(2 - \sqrt{3})$

26) a)

$\sinh^2 x + \sinh x - 6 = 0$

$\Rightarrow (\sinh x + 3)(\sinh x - 2) = 0$

$\Rightarrow \sinh x = -3$ or $\sinh x = 2$

First consider $\sinh x = -3$:

$\Rightarrow \frac{e^x - e^{-x}}{2} = -3$

$\Rightarrow e^x - e^{-x} = -6$

$\Rightarrow e^{2x} - 1 = -6e^x$

$\Rightarrow e^{2x} + 6e^x - 1 = 0$

$\Rightarrow (e^x + 3)^2 - 3^2 - 1 = 0$

$\Rightarrow (e^x + 3)^2 = 10$

$\Rightarrow e^x = -3 \pm \sqrt{10}$

$\Rightarrow x = \ln(-3 + \sqrt{10})$ or $x = \ln(-3 - \sqrt{10})$

However, $-3 - \sqrt{10} < 0$, so the only real solution is $x = \ln(-3 + \sqrt{10})$

Now consider $\sinh x = 2$:

$\Rightarrow \frac{e^x - e^{-x}}{2} = 2$

$\Rightarrow e^x - e^{-x} = 4$

$\Rightarrow e^{2x} - 1 = 4e^x$

$\Rightarrow e^{2x} - 4e^x - 1 = 0$

$\Rightarrow (e^x - 2)^2 - 2^2 - 1 = 0$

$\Rightarrow (e^x - 2)^2 = 5$

$\Rightarrow e^x = 2 \pm \sqrt{5}$

$\Rightarrow x = \ln(2 + \sqrt{5})$ or $x = \ln(2 - \sqrt{5})$

However, $2 - \sqrt{5} < 0$, so the only real solution is $x = \ln(2 + \sqrt{5})$

So $x = \ln(-3 + \sqrt{10})$ and $x = \ln(2 + \sqrt{5})$ solve the original equation.

26) b)

$8\cosh^2 x - 26\cosh x + 15 = 0$

$\Rightarrow (4\cosh x - 3)(2\cosh x - 5) = 0$

$\Rightarrow \cosh x = \frac{3}{4}$ or $\cosh x = \frac{5}{2}$

First consider $\cosh x = \frac{3}{4}$:

$\cosh x \geq 1$ for all real values of x, so $\cosh x = \frac{3}{4}$ has no real solutions.

Now consider $\cosh x = \frac{5}{2}$:

$\Rightarrow \frac{e^x + e^{-x}}{2} = \frac{5}{2}$

$\Rightarrow e^x + e^{-x} = 5$

$\Rightarrow e^{2x} + 1 = 5e^x$

$\Rightarrow e^{2x} - 5e^x + 1 = 0$

$\Rightarrow \left(e^x - \frac{5}{2}\right)^2 - \left(\frac{5}{2}\right)^2 + 1 = 0$

$\Rightarrow \left(e^x - \frac{5}{2}\right)^2 = \frac{21}{4}$

$\Rightarrow e^x = \frac{5}{2} \pm \frac{\sqrt{21}}{2}$

$\Rightarrow x = \ln\left(\frac{5}{2} + \frac{\sqrt{21}}{2}\right)$ or $x = \ln\left(\frac{5}{2} - \frac{\sqrt{21}}{2}\right)$

27) a)

$\cosh^3 x - 5\cosh^2 x - 2\cosh x + 24 = 0$

$\Rightarrow (\cosh x + 2)(\cosh x - 3)(\cosh x - 4) = 0$

$\Rightarrow \cosh x = -2$ or $\cosh x = 3$ or $\cosh x = 4$

First consider $\cosh x = -2$:

$\cosh x \geq 1$ for all real values of x, so $\cosh x = -2$ has no real solutions.

Now consider $\cosh x = 3$:

$\Rightarrow \frac{e^x + e^{-x}}{2} = 3$

$\Rightarrow e^x + e^{-x} = 6$

$\Rightarrow e^{2x} + 1 = 6e^x$

$\Rightarrow e^{2x} - 6e^x + 1 = 0$

$\Rightarrow (e^x - 3)^2 - 3^2 + 1 = 0$

$\Rightarrow (e^x - 3)^2 = 8$

$\Rightarrow e^x = 3 \pm 2\sqrt{2}$

$\Rightarrow x = \ln(3 + 2\sqrt{2})$ or $x = \ln(3 - 2\sqrt{2})$

Now consider $\cosh x = 4$:

$\Rightarrow \frac{e^x + e^{-x}}{2} = 4$

$\Rightarrow e^x + e^{-x} = 8$

$\Rightarrow e^{2x} + 1 = 8e^x$

$\Rightarrow e^{2x} - 8e^x + 1 = 0$

$\Rightarrow (e^x - 4)^2 - 4^2 + 1 = 0$

$\Rightarrow (e^x - 4)^2 = 15$

$\Rightarrow e^x = 4 \pm \sqrt{15}$

$\Rightarrow x = \ln(4 + \sqrt{15})$ or $x = \ln(4 - \sqrt{15})$

So $x = \ln(3 \pm 2\sqrt{2})$ and $x = \ln(4 \pm \sqrt{15})$ solve the original equation.

27) b)

$12\cosh^4 x - 44\cosh^2 x - 45 = 0$

$\Rightarrow (6\cosh^2 x + 5)(2\cosh^2 x - 9) = 0$

$\Rightarrow \cosh^2 x = -\frac{5}{6}$ or $\cosh^2 x = \frac{9}{2}$

Firstly, $\cosh^2 x = -\frac{5}{6}$ cannot have any real solutions.

Now consider $\cosh^2 x = \frac{9}{2} \Rightarrow \cosh x = \pm\frac{3\sqrt{2}}{2}$

First consider $\cosh x = -\frac{3\sqrt{2}}{2}$:

$\cosh x \geq 1$ for all real values of x, so $\cosh x = -\frac{3\sqrt{2}}{2}$ has no real solutions

Now consider $\cosh x = \frac{3\sqrt{2}}{2}$:

$\Rightarrow \frac{e^x + e^{-x}}{2} = \frac{3\sqrt{2}}{2}$

$\Rightarrow e^x + e^{-x} = 3\sqrt{2}$

$\Rightarrow e^{2x} + 1 = 3\sqrt{2}e^x$

$\Rightarrow e^{2x} - 3\sqrt{2}e^x + 1 = 0$

$\Rightarrow \left(e^x - \frac{3\sqrt{2}}{2}\right)^2 - \left(\frac{3\sqrt{2}}{2}\right)^2 + 1 = 0$

$\Rightarrow \left(e^x - \frac{3\sqrt{2}}{2}\right)^2 = \frac{7}{2}$

$\Rightarrow e^x = \frac{3\sqrt{2}}{2} \pm \frac{\sqrt{14}}{2}$

$\Rightarrow x = \ln\left(\frac{3\sqrt{2}}{2} + \frac{\sqrt{14}}{2}\right)$ or $x = \ln\left(\frac{3\sqrt{2}}{2} - \frac{\sqrt{14}}{2}\right)$

28) a)

$15\tanh^2 x - 8\tanh x + 1 = 0$

$\Rightarrow (5\tanh x - 1)(3\tanh x - 1) = 0$

$\Rightarrow \tanh x = \frac{1}{5}$ or $\tanh x = \frac{1}{3}$

First consider $\tanh x = \frac{1}{5}$:

$\Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = \frac{1}{5}$

$\Rightarrow 5(e^{2x} - 1) = e^{2x} + 1$

$\Rightarrow 5e^{2x} - 5 = e^{2x} + 1$

$\Rightarrow 4e^{2x} = 6$

$\Rightarrow e^{2x} = \frac{3}{2}$

$\Rightarrow 2x = \ln\frac{3}{2}$

$\Rightarrow x = \frac{1}{2}\ln\frac{3}{2}$

Now consider $\tanh x = \frac{1}{3}$:

$\Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = \frac{1}{3}$

$\Rightarrow 3(e^{2x} - 1) = e^{2x} + 1$

$\Rightarrow 3e^{2x} - 3 = e^{2x} + 1$

$\Rightarrow 2e^{2x} = 4$

$\Rightarrow e^{2x} = 2$

$\Rightarrow 2x = \ln 2$

$\Rightarrow x = \frac{1}{2}\ln 2$

So $x = \frac{1}{2}\ln\frac{3}{2}$ and $x = \frac{1}{2}\ln 2$ solve the original equation.

28) b)

$12\tanh^2 x - 17\tanh x + 6 = 0$

$\Rightarrow (3\tanh x - 2)(4\tanh x - 3) = 0$

$\Rightarrow \tanh x = \frac{2}{3}$ or $\tanh x = \frac{3}{4}$

First consider $\tanh x = \frac{2}{3}$:

$\Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = \frac{2}{3}$

$\Rightarrow 3(e^{2x} - 1) = 2(e^{2x} + 1)$

$\Rightarrow 3e^{2x} - 3 = 2e^{2x} + 2$

$\Rightarrow e^{2x} = 5$

$\Rightarrow 2x = \ln 5$

$\Rightarrow x = \frac{1}{2}\ln 5$

Now consider $\tanh x = \frac{3}{4}$:

$\Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = \frac{3}{4}$

$\Rightarrow 4(e^{2x} - 1) = 3(e^{2x} + 1)$

$\Rightarrow 4e^{2x} - 4 = 3e^{2x} + 3$

$\Rightarrow e^{2x} = 7$

$\Rightarrow 2x = \ln 7$

$\Rightarrow x = \frac{1}{2}\ln 7$

So $x = \frac{1}{2}\ln 5$ and $x = \frac{1}{2}\ln 7$ solve the original equation.

29) a) i)

$f(1) = 1^3 - 7 \times 1^2 + 7 \times 1 - 1 = 1 - 7 + +7 - 1 = 0$

$\therefore$ both $(y - 1)$ is a factor of $f(y)$ by the Factor Theorem.

29) a) ii)

$f(y) = y^3 - 7y^3 + 7y - 1$

$\equiv (y - 1)(y^2 - 6y + 1)$

29) b)

$3\tanh 2x = 4\tanh x$

$\Rightarrow 3 \times \frac{e^{2(2x)}-1}{e^{2(2x)}+1} = 4 \times \frac{e^{2x}-1}{e^{2x}+1}$

$\Rightarrow 3(e^{4x}-1)(e^{2x}+1) = 4(e^{4x}+1)(e^{2x}-1)$

$\Rightarrow 3(e^{6x}+e^{4x}-e^{2x}-1) = 4(e^{6x}-e^{4x}+e^{2x}-1)$

$\Rightarrow 3e^{6x}+3e^{4x}-3e^{2x}-3 = 4e^{6x}-4e^{4x}+4e^{2x}-4$

$\Rightarrow 0 = e^{6x}-7e^{4x}+7e^{2x}-1$

$\Rightarrow (e^{2x}-1)(e^{4x}-6e^{2x}+1) = 0$

$\Rightarrow e^{2x} = 1$ or $e^{4x}-6e^{2x}+1 = 0$

First consider $e^{2x} = 1$:

$\Rightarrow 2x = 0 \Rightarrow x = 0$

Now consider $e^{4x}-6e^{2x}+1 = 0$

$\Rightarrow (e^{2x}-3)^2 - 3^2 + 1 = 0$

$\Rightarrow (e^{2x}-3)^2 = 8$

$\Rightarrow e^{2x} = 3 \pm 2\sqrt{2}$

$\Rightarrow x = \frac{1}{2}\ln(3+2\sqrt{2})$ or $x = \frac{1}{2}\ln(3-2\sqrt{2})$

So $x = 0$, $x = \frac{1}{2}\ln(3+2\sqrt{2})$ and $x = \frac{1}{2}\ln(3-2\sqrt{2})$ solve the original equation.

$x = \frac{1}{2}\ln(3+2\sqrt{2}) = \ln\left((3+2\sqrt{2})^{\frac{1}{2}}\right)$

$(a+b\sqrt{2})^2 = 3+2\sqrt{2}$, where a and b are rational numbers.

$\Rightarrow a^2+2b^2+2ab\sqrt{2} = 3+2\sqrt{2}$

$\therefore a^2+2b^2 = 3$ (1)

$ab = 1 \Rightarrow b = \frac{1}{a}$ (2)

Substitute (2) into (1):

$a^2+2\left(\frac{1}{a}\right)^2 = 3 \Rightarrow a^4+2 = 3a^2 \Rightarrow a^4-3a^2+2 = 0 \Rightarrow (a^2-2)(a^2-1) = 0$

$a^2-2 = 0 \Rightarrow a = \pm\sqrt{2}$ but a is a rational number, so this is not solution.

$a^2-1 = 0 \Rightarrow a = \pm 1$

When $a = 1$, $b = \frac{1}{1} = 1$

When $a = -1$, $b = \frac{1}{-1} = -1$

$(3+2\sqrt{2})^{\frac{1}{2}} > 0$ so $(3+2\sqrt{2})^{\frac{1}{2}} = 1+\sqrt{2}$

$x = \frac{1}{2}\ln\left(3 - 2\sqrt{2}\right) = \ln\left(\left(3 - 2\sqrt{2}\right)^{\frac{1}{2}}\right)$

$\left(a + b\sqrt{2}\right)^2 = 3 - 2\sqrt{2}$, where a and b are rational numbers.

$\Rightarrow a^2 + 2b^2 + 2ab\sqrt{2} = 3 - 2\sqrt{2}$

$\therefore a^2 + 2b^2 = 3$ (1)

$ab = -1 \Rightarrow b = -\frac{1}{a}$ (2)

Substitute (2) into (1):

$a^2 + 2\left(-\frac{1}{a}\right)^2 = 3 \Rightarrow a^4 + 2 = 3a^2 \Rightarrow a^4 - 3a^2 + 2 = 0 \Rightarrow (a^2 - 2)(a^2 - 1) = 0$

$a^2 - 2 = 0 \Rightarrow a = \pm\sqrt{2}$ but a is a rational number, so this is not solution.

$a^2 - 1 = 0 \Rightarrow a = \pm 1$

When $a = 1$, $b = -\frac{1}{1} = -1$

When $a = -1$, $b = -\frac{1}{-1} = 1$

$\ln\left(1 - \sqrt{2}\right)$ is undefined so $\left(3 - 2\sqrt{2}\right)^{\frac{1}{2}} = -1 + \sqrt{2}$ is the only solution in this case.

So $x = 0$, $x = \frac{1}{2}\ln\left(3 + 2\sqrt{2}\right) = \ln\left(1 + \sqrt{2}\right)$ and $x = \frac{1}{2}\ln\left(3 - 2\sqrt{2}\right) = \ln\left(\sqrt{2} - 1\right)$ are the three real solutions of $3\tanh 2x = 4\tanh x$

30) a)

$\frac{1}{\cosh x} + \frac{1}{\sinh x} = \mathrm{e}^{3x}$

$\Rightarrow \frac{\sinh x + \cosh x}{\sinh x \cosh x} = \mathrm{e}^{3x}$

$\Rightarrow \sinh x + \cosh x = \mathrm{e}^{3x}\sinh x \cosh x$

$\Rightarrow \frac{\mathrm{e}^x - \mathrm{e}^{-x}}{2} + \frac{\mathrm{e}^x + \mathrm{e}^{-x}}{2} = \mathrm{e}^{3x} \times \frac{\mathrm{e}^x - \mathrm{e}^{-x}}{2} \times \frac{\mathrm{e}^x + \mathrm{e}^{-x}}{2}$

$\Rightarrow 2(\mathrm{e}^x - \mathrm{e}^{-x}) + 2(\mathrm{e}^x + \mathrm{e}^{-x}) = \mathrm{e}^{3x}(\mathrm{e}^x - \mathrm{e}^{-x})(\mathrm{e}^x + \mathrm{e}^{-x})$

$\Rightarrow 2\mathrm{e}^x - 2\mathrm{e}^{-x} + 2\mathrm{e}^x + 2\mathrm{e}^{-x} = \mathrm{e}^{3x}(\mathrm{e}^{2x} - \mathrm{e}^{-2x})$

$\Rightarrow 4\mathrm{e}^x = \mathrm{e}^{5x} - \mathrm{e}^x$

$\Rightarrow \mathrm{e}^{5x} = 5\mathrm{e}^x$

$\Rightarrow \mathrm{e}^{4x} = 5$

$\Rightarrow 4x = \ln 5$

$\Rightarrow x = \frac{1}{4}\ln 5$

30) b)

$$\frac{1}{\cosh x}-\frac{1}{\sinh x}=\frac{4}{e^{3x}}$$
$$\Rightarrow \frac{\sinh x-\cosh x}{\sinh x\cosh x}=\frac{4}{e^{3x}}$$
$$\Rightarrow e^{3x}(\sinh x-\cosh x)=4\sinh x\cosh x$$
$$\Rightarrow e^{3x}\left(\frac{e^{x}-e^{-x}}{2}-\frac{e^{x}+e^{-x}}{2}\right)=4\times\frac{e^{x}-e^{-x}}{2}\times\frac{e^{x}+e^{-x}}{2}$$
$$\Rightarrow 2e^{3x}\left(e^{x}-e^{-x}-(e^{x}+e^{-x})\right)=4(e^{x}-e^{-x})(e^{x}+e^{-x})$$
$$\Rightarrow e^{3x}(-2e^{-x})=2(e^{2x}-e^{-2x})$$
$$\Rightarrow -2e^{2x}=2e^{2x}-2e^{-2x}$$
$$\Rightarrow -2e^{4x}=2e^{4x}-2$$
$$\Rightarrow 4e^{4x}=2$$
$$\Rightarrow e^{4x}=\frac{1}{2}$$
$$\Rightarrow 4x=\ln\frac{1}{2}$$
$$\Rightarrow x=\frac{1}{4}\ln\frac{1}{2}$$

30) c)

$$\frac{1}{\cosh x}+\frac{\sinh x}{1}=\frac{2}{e^{x}}$$
$$\Rightarrow \frac{1+\sinh x\cosh x}{\cosh x}=\frac{2}{e^{x}}$$
$$\Rightarrow e^{x}(1+\sinh x\cosh x)=2\cosh x$$
$$\Rightarrow e^{x}\left(1+\frac{e^{x}-e^{-x}}{2}\times\frac{e^{x}+e^{-x}}{2}\right)=2\times\frac{e^{x}+e^{-x}}{2}$$
$$\Rightarrow e^{x}\left(4+(e^{x}-e^{-x})(e^{x}+e^{-x})\right)=4(e^{x}+e^{-x})$$
$$\Rightarrow e^{x}(4+e^{2x}-e^{-2x})=4e^{x}+4e^{-x}$$
$$\Rightarrow 4e^{x}+e^{3x}-e^{-x}=4e^{x}+4e^{-x}$$
$$\Rightarrow e^{3x}=5e^{-x}$$
$$\Rightarrow e^{4x}=5$$
$$\Rightarrow 4x=\ln 5$$
$$\Rightarrow x=\frac{1}{4}\ln 5$$

30) d)

$\frac{4\cosh x}{1} - \frac{1}{\sinh x} = \frac{2}{e^x}$

$\Rightarrow \frac{4\sinh x \cosh x - 1}{\sinh x} = \frac{2}{e^x}$

$\Rightarrow e^x(4\sinh x \cosh x - 1) = 2\sinh x$

$\Rightarrow e^x\left(4 \times \frac{e^x - e^{-x}}{2} \times \frac{e^x + e^{-x}}{2} - 1\right) = 2 \times \frac{e^x - e^{-x}}{2}$

$\Rightarrow e^x\left((e^x - e^{-x})(e^x + e^{-x}) - 1\right) = e^x - e^{-x}$

$\Rightarrow e^x(e^{2x} - e^{-2x} - 1) = e^x - e^{-x}$

$\Rightarrow e^{3x} - e^{-x} - e^x = e^x - e^{-x}$

$\Rightarrow e^{3x} - 2e^x = 0$

$\Rightarrow e^x(e^{2x} - 2) = 0$

$e^x \neq 0$ so $e^{2x} - 2 = 0$

$\Rightarrow e^{2x} = 2$

$\Rightarrow 2x = \ln 2$

$\Rightarrow x = \frac{1}{2}\ln 2$

31) a) i)

$f(-1) = (-1)^4 + 8 \times (-1)^3 + 2 \times (-1)^2 - 16 \times (-1) - 11$

$= 1 - 8 + 2 + 16 - 11 = 0$

$\therefore (y + 1)$ is a factor of $f(y)$ by the Factor Theorem.

31) a) ii)

$f(y) = y^4 + 8y^3 + 2y^2 - 16y - 11$

$\equiv (y + 1)(y^3 + 7y^2 - 5y - 11)$

$\equiv (y + 1)(y + 1)(y^2 + 6y - 11)$

31) b)

$\frac{2\sinh x}{1+\cosh x}+\frac{1+\cosh x}{2\sinh x}=3$

$\Rightarrow \frac{4\sinh^2 x+(1+\cosh x)^2}{2\sinh x(1+\cosh x)}=3$

$\Rightarrow 4\sinh^2 x+(1+\cosh x)^2=6\sinh x\,(1+\cosh x)$

$\Rightarrow 4\sinh^2 x+1+2\cosh x+\cosh^2 x=6\sinh x+6\sinh x\cosh x$

$\Rightarrow 4\times\left(\frac{e^x-e^{-x}}{2}\right)^2+1+2\times\frac{e^x+e^{-x}}{2}+\left(\frac{e^x+e^{-x}}{2}\right)^2=6\times\frac{e^x-e^{-x}}{2}+6\times\frac{e^x-e^{-x}}{2}\times\frac{e^x+e^{-x}}{2}$

$\Rightarrow (e^x-e^{-x})^2+1+e^x+e^{-x}+\frac{1}{4}(e^x+e^{-x})^2=3(e^x-e^{-x})+\frac{3}{2}(e^x-e^{-x})(e^x+e^{-x})$

$\Rightarrow e^{2x}-2+e^{-2x}+1+e^x+e^{-x}+\frac{1}{4}(e^{2x}+2+e^{-2x})=3e^x-3e^{-x}+\frac{3}{2}(e^{2x}-e^{-2x})$

$\Rightarrow 4e^{2x}-8+4e^{-2x}+4+4e^x+4e^{-x}+e^{2x}+2+e^{-2x}=12e^x-12e^{-x}+6e^{2x}-6e^{-2x}$

$\Rightarrow -2=8e^x-16e^{-x}+e^{2x}-11e^{-2x}$

$\Rightarrow -2e^{2x}=8e^{3x}-16e^x+e^{4x}-11$

$\Rightarrow e^{4x}+8e^{3x}+2e^{2x}-16e^x-11=0$

$\Rightarrow (e^x+1)^2(e^{2x}+6e^x-11)=0$

$\Rightarrow e^x+1=0$ or $e^{2x}+6e^x-11=0$

First consider $e^x+1=0$:

$e^x>0$ so this has no real solutions.

Now consider $e^{2x}+6e^x-11=0$:

$(e^x+3)^2-3^2-11=0$

$\Rightarrow (e^x+3)^2=20$

$\Rightarrow e^x=-3\pm 2\sqrt{5}$

$\Rightarrow x=\ln\left(-3+2\sqrt{5}\right)$ or $x=\ln\left(-3-2\sqrt{5}\right)$

However, $-3-2\sqrt{5}<0$, so the only real solution is $x=\ln\left(-3+2\sqrt{5}\right)$

32)

$\cosh^3 x+\sinh 3x$

$\equiv\left(\frac{e^x+e^{-x}}{2}\right)^3+\frac{e^{3x}-e^{-3x}}{2}$

$\equiv\frac{1}{8}(e^x+e^{-x})^3+\frac{1}{2}(e^{3x}-e^{-3x})$

$\equiv\frac{1}{8}(e^{3x}+3e^x+3e^{-x}+e^{-3x})+\frac{1}{8}(4e^{3x}-4e^{-3x})$

$\equiv\frac{1}{8}(e^{3x}+3e^x+3e^{-x}+e^{-3x}+4e^{3x}-4e^{-3x})$

$\equiv\frac{1}{8}(5e^{3x}+3e^x+3e^{-x}-3e^{-3x})$

33) a) i)

$4\sinh^3 x - 4\cosh^3 x$

$\equiv 4\left(\frac{e^x - e^{-x}}{2}\right)^3 - 4\left(\frac{e^x + e^{-x}}{2}\right)^3$

$\equiv \frac{1}{2}(e^x - e^{-x})^3 - \frac{1}{2}(e^x + e^{-x})^3$

$\equiv \frac{1}{2}(e^{3x} - 3e^x + 3e^{-x} - e^{-3x}) - \frac{1}{2}(e^{3x} + 3e^x + 3e^{-x} + e^{-3x})$

$\equiv \frac{1}{2}\left(e^{3x} - 3e^x + 3e^{-x} - e^{-3x} - (e^{3x} + 3e^x + 3e^{-x} + e^{-3x})\right)$

$\equiv \frac{1}{2}(e^{3x} - 3e^x + 3e^{-x} - e^{-3x} - e^{3x} - 3e^x - 3e^{-x} - e^{-3x})$

$\equiv \frac{1}{2}(-6e^x - 2e^{-3x})$

$\equiv -(e^{-3x} + 3e^x)$

33) a) ii)

$4\sinh^3 x + 4\cosh^3 x$

$\equiv 4\left(\frac{e^x - e^{-x}}{2}\right)^3 + 4\left(\frac{e^x + e^{-x}}{2}\right)^3$

$\equiv \frac{1}{2}(e^x - e^{-x})^3 + \frac{1}{2}(e^x + e^{-x})^3$

$\equiv \frac{1}{2}(e^{3x} - 3e^x + 3e^{-x} - e^{-3x}) + \frac{1}{2}(e^{3x} + 3e^x + 3e^{-x} + e^{-3x})$

$\equiv \frac{1}{2}(e^{3x} - 3e^x + 3e^{-x} - e^{-3x} + e^{3x} + 3e^x + 3e^{-x} + e^{-3x})$

$\equiv \frac{1}{2}(2e^{3x} + 6e^{-x})$

$\equiv e^{3x} + 3e^{-x}$

33) b)

$16\sinh^6 x - 16\cosh^6 x$

$\equiv (4\sinh^3 x + 4\cosh^3 x)(4\sinh^3 x - 4\cosh^3 x)$

$\equiv (e^{3x} + 3e^{-x})\left(-(e^{-3x} + 3e^x)\right)$

$\equiv -(e^{3x} + 3e^{-x})(e^{-3x} + 3e^x)$

$\equiv -(1 + 3e^{4x} + 3e^{-4x} + 3)$

$\equiv -(3(e^{4x} + e^{-4x}) + 4)$

$\equiv -\left(6 \times \frac{e^{4x} + e^{-4x}}{2} + 4\right)$

$\equiv -2(3\cosh 4x + 2)$

34)

LHS: $(\sinh x + \cosh x)^3 \equiv \left(\frac{e^x - e^{-x}}{2} + \frac{e^x + e^{-x}}{2}\right)^3$

$\equiv \left(\frac{e^x - e^{-x} + e^x + e^{-x}}{2}\right)^3$

$\equiv \left(\frac{2e^x}{2}\right)^3 \equiv (e^x)^3 = e^{3x}$

35) a)

LHS: $4\cosh^3 x \equiv 4\left(\frac{e^x + e^{-x}}{2}\right)^3$

$\equiv 4 \times \frac{(e^x + e^{-x})^3}{8}$

$\equiv \frac{1}{2}(e^x + e^{-x})^3$

$\equiv \frac{1}{2}(e^{3x} + 3e^x + 3e^{-x} + e^{-3x})$

$\equiv \frac{e^{3x} + e^{-3x}}{2} + 3 \times \frac{e^x + e^{-x}}{2}$

$\equiv \cosh 3x + 3\cosh x$: RHS

35) b)

Let $y = \cosh x$

$4\cosh^3 x = \cosh 3x + 3\cosh x \Rightarrow 4y^3 = \cosh 3x + 3y$

$\Rightarrow 4y^3 - 3y = \cosh 3x$

$\cosh 3x = \frac{5}{4}$

$\Rightarrow \frac{e^{3x}+e^{-3x}}{2} = \frac{5}{4}$

$\Rightarrow e^{3x} + e^{-3x} = \frac{5}{2}$

$\Rightarrow e^{6x} + 1 = \frac{5}{2}e^{3x}$

$\Rightarrow e^{6x} - \frac{5}{2}e^{3x} + 1 = 0$

$\Rightarrow \left(e^{3x} - \frac{5}{4}\right)^2 - \left(\frac{5}{4}\right)^2 + 1 = 0$

$\Rightarrow \left(e^{3x} - \frac{5}{4}\right)^2 = \frac{9}{16}$

$\Rightarrow e^{3x} = \frac{5}{4} \pm \frac{3}{4}$

$\Rightarrow x = \frac{1}{3}\ln\left(\frac{5}{4} \pm \frac{3}{4}\right)$

$\Rightarrow x = \frac{1}{3}\ln 2$ or $x = \frac{1}{3}\ln\frac{1}{2}$

So $y = \cosh\left(\frac{1}{3}\ln 2\right)$ or $y = \cosh\left(\frac{1}{3}\ln\frac{1}{2}\right)$

Consider $y = \cosh\left(\frac{1}{3}\ln 2\right)$ first:

$\Rightarrow y = \frac{e^{\frac{1}{3}\ln 2}+e^{-\frac{1}{3}\ln 2}}{2}$

$\Rightarrow y = \frac{e^{\ln 2^{\frac{1}{3}}}+e^{\ln 2^{-\frac{1}{3}}}}{2}$

$\Rightarrow y = \frac{2^{\frac{1}{3}}+2^{-\frac{1}{3}}}{2}$

$\Rightarrow y = 2^{-\frac{2}{3}} + 2^{-\frac{4}{3}}$

Now consider $y = \cosh\left(\frac{1}{3}\ln\frac{1}{2}\right)$:

$\Rightarrow y = \frac{e^{\frac{1}{3}\ln\frac{1}{2}}+e^{-\frac{1}{3}\ln\frac{1}{2}}}{2}$

$\Rightarrow y = \frac{e^{\frac{1}{3}\ln 2^{-1}}+e^{-\frac{1}{3}\ln 2^{-1}}}{2}$

$\Rightarrow y = \frac{e^{\ln 2^{-\frac{1}{3}}}+e^{\ln 2^{\frac{1}{3}}}}{2}$

$\Rightarrow y = \frac{2^{-\frac{1}{3}} + 2^{\frac{1}{3}}}{2}$

$\Rightarrow y = 2^{-\frac{4}{3}} + 2^{-\frac{2}{3}}$ which is the same result as before.

36) a)

$\cosh x \tanh x \equiv \cosh x \times \frac{\sinh x}{\cosh x} \equiv \sinh x$

36) b)

$\frac{\tanh x}{\sinh x} \equiv \frac{\frac{\sinh x}{\cosh x}}{\sinh x} \equiv \frac{1}{\cosh x}$

35) c)

$\frac{\tanh x}{\cosh x} \equiv \frac{\frac{\sinh x}{\cosh x}}{\cosh x} \equiv \frac{\sinh x}{\cosh^2 x}$

37) a)

$\sinh x \cosh x \tanh x \equiv \sinh x \cosh x \times \frac{\sinh x}{\cosh x} \equiv \sinh^2 x$

37) b)

$\sinh^2 x \cosh x \tanh^2 x \equiv \sinh^2 x \cosh x \times \frac{\sinh^2 x}{\cosh^2 x} \equiv \frac{\sinh^4 x}{\cosh x}$

37) c)

$\frac{\sinh x}{\cosh^2 x \tanh x} \equiv \frac{\sinh x}{\cosh^2 x \times \frac{\sinh x}{\cosh x}} \equiv \frac{\sinh x}{\cosh x \sinh x} \equiv \frac{1}{\cosh x}$

37) d)

$\frac{\tanh^2 x}{\sinh^2 x \cosh x} \equiv \frac{\frac{\sinh^2 x}{\cosh^2 x}}{\sinh^2 x \cosh x} \equiv \frac{\frac{1}{\cosh^2 x}}{\cosh x} \equiv \frac{1}{\cosh^3 x}$

38)

LHS: $\cosh^2 x - \sinh^2 x \equiv \left(\frac{e^x+e^{-x}}{2}\right)^2 - \left(\frac{e^x-e^{-x}}{2}\right)^2$

$\equiv \frac{(e^x+e^{-x})^2}{4} - \frac{(e^x-e^{-x})^2}{4}$

$\equiv \frac{e^{2x}+2+e^{-2x}}{4} - \frac{e^{2x}-2+e^{-2x}}{4}$

$\equiv \frac{e^{2x}+2+e^{-2x}-(e^{2x}-2+e^{-2x})}{4}$

$\equiv \frac{e^{2x}+2+e^{-2x}-e^{2x}+2-e^{-2x}}{4}$

$\equiv \frac{4}{4} \equiv 1$: RHS

39)

RHS: $2\sinh x \cosh x \equiv 2 \times \frac{e^x-e^{-x}}{2} \times \frac{e^x+e^{-x}}{2}$

$\equiv \frac{(e^x-e^{-x})(e^x+e^{-x})}{2}$

$\equiv \frac{e^{2x}-e^{-2x}}{2} \equiv \sinh 2x$: LHS

40) a)

RHS: $\cosh^2 x + \sinh^2 x \equiv \left(\frac{e^x+e^{-x}}{2}\right)^2 + \left(\frac{e^x-e^{-x}}{2}\right)^2$

$\equiv \frac{(e^x+e^{-x})^2}{4} + \frac{(e^x-e^{-x})^2}{4}$

$\equiv \frac{e^{2x}+2+e^{-2x}}{4} + \frac{e^{2x}-2+e^{-2x}}{4}$

$\equiv \frac{e^{2x}+2+e^{-2x}+e^{2x}-2+e^{-2x}}{4}$

$\equiv \frac{2e^{2x}+2e^{-2x}}{4}$

$\equiv \frac{e^{2x}+e^{-2x}}{2} \equiv \cosh 2x$:LHS

40) b)

RHS: $2\cosh^2 x - 1 \equiv 2 \times \left(\frac{e^x + e^{-x}}{2}\right)^2 - 1$

$\equiv 2 \times \frac{(e^x + e^{-x})^2}{4} - 1$

$\equiv \frac{e^{2x} + 2 + e^{-2x}}{2} - 1$

$\equiv \frac{e^{2x} + e^{-2x}}{2} + \frac{2}{2} - 1$

$\equiv \frac{e^{2x} + e^{-2x}}{2} \equiv \cosh 2x$:LHS

40) c)

RHS: $1 + 2\sinh^2 x \equiv 1 + 2 \times \left(\frac{e^x - e^{-x}}{2}\right)^2$

$\equiv 1 + 2 \times \frac{(e^x - e^{-x})^2}{4}$

$\equiv 1 + \frac{e^{2x} - 2 + e^{-2x}}{2}$

$\equiv 1 - \frac{2}{2} + \frac{e^{2x} + e^{-2x}}{2}$

$\equiv \frac{e^{2x} + e^{-2x}}{2} \equiv \cosh 2x$:LHS

41)

RHS: $\frac{2\tanh x}{1 + \tanh^2 x} \equiv \frac{2 \times \frac{e^{2x} - 1}{e^{2x} + 1}}{1 + \left(\frac{e^{2x} - 1}{e^{2x} + 1}\right)^2}$

$\equiv \frac{2 \times \frac{e^{2x} - 1}{e^{2x} + 1}}{1 + \frac{(e^{2x} - 1)^2}{(e^{2x} + 1)^2}}$

$\equiv \frac{2(e^{2x} - 1)(e^{2x} + 1)}{(e^{2x} + 1)^2 + (e^{2x} - 1)^2}$

$\equiv \frac{2(e^{4x} - 1)}{e^{4x} + 2e^{2x} + 1 + e^{4x} - 2e^{2x} + 1}$

$\equiv \frac{2e^{4x} - 2}{2e^{4x} + 2}$

$\equiv \frac{e^{4x} - 1}{e^{4x} + 1}$

$\equiv \frac{e^{2(2x)} - 1}{e^{2(2x)} + 1} \equiv \tanh 2x$: LHS

42) a)

$5\cosh^2 x + 3\sinh^2 x \equiv 5(1 + \sinh^2 x) + 3\sinh^2 x$

$\equiv 5 + 5\sinh^2 x + 3\sinh^2 x$

$\equiv 5 + 8\sinh^2 x$

42) b)

$2\sinh^2 x - 6\cosh^2 x \equiv 2\sinh^2 x - 6(1 + \sinh^2 x)$

$\equiv 2\sinh^2 x - 6 - 6\sinh^2 x$

$\equiv -4\sinh^2 x - 6$

43) a)

$9\cosh^2 x + 2\sinh^2 x \equiv 9\cosh^2 x + 2(\cosh^2 x - 1)$

$\equiv 9\cosh^2 x + 2\cosh^2 x - 2$

$\equiv 11\cosh^2 x - 2$

43) b)

$5\sinh^2 x - 8\cosh^2 x \equiv 5(\cosh^2 x - 1) - 8\cosh^2 x$

$\equiv 5\cosh^2 x - 5 - 8\cosh^2 x$

$\equiv -3\cosh^2 x - 5$

44) a)

$\cosh 2x \equiv 2\cosh^2 x - 1 \Rightarrow \cosh^2 x \equiv \frac{\cosh 2x + 1}{2}$

$\cosh 2x \equiv 1 + 2\sinh^2 x \Rightarrow \sinh^2 x \equiv \frac{\cosh 2x - 1}{2}$

So:

$2\cosh^2 x + 5\sinh^2 x \equiv 2 \times \frac{\cosh 2x + 1}{2} + 5 \times \frac{\cosh 2x - 1}{2}$

$\equiv \cosh 2x + 1 + \frac{5}{2}\cosh 2x - \frac{5}{2}$

$\equiv \frac{7}{2}\cosh 2x - \frac{3}{2}$

44) b)

$8\sinh^2 x - 3\cosh^2 x \equiv 8 \times \frac{\cosh 2x - 1}{2} - 3 \times \frac{\cosh 2x + 1}{2}$

$\equiv 4(\cosh 2x - 1) - \frac{3}{2}(\cosh 2x + 1)$

$\equiv 4\cosh 2x - 4 - \frac{3}{2}\cosh 2x - \frac{3}{2}$

$\equiv \frac{5}{2}\cosh 2x - \frac{11}{2}$

45) a)

$\cosh^2 x + \sinh^2 x = 3$

$\Rightarrow \cosh^2 x + \cosh^2 x - 1 = 3$

$\Rightarrow 2\cosh^2 x = 4$

$\Rightarrow \cosh^2 x = 2$

$\Rightarrow \cosh x = \pm\sqrt{2}$

Consider $\cosh x = -\sqrt{2}$ first:

$\cosh x \geq 1$ so the equation $\cosh x = -\sqrt{2}$ has no real solutions.

Now consider $\cosh x = \sqrt{2}$:

$\Rightarrow \frac{e^x + e^{-x}}{2} = \sqrt{2}$

$\Rightarrow e^x + e^{-x} = 2\sqrt{2}$

$\Rightarrow e^{2x} + 1 = 2\sqrt{2}e^x$

$\Rightarrow e^{2x} - 2\sqrt{2}e^x + 1 = 0$

$\Rightarrow \left(e^x - \sqrt{2}\right)^2 - \left(\sqrt{2}\right)^2 + 1 = 0$

$\Rightarrow \left(e^x - \sqrt{2}\right)^2 = 1$

$\Rightarrow e^x = \sqrt{2} \pm 1$

$\Rightarrow x = \ln\left(\sqrt{2} \pm 1\right)$

$\Rightarrow x = \ln\left(\sqrt{2} + 1\right)$ or $x = \ln\left(\sqrt{2} - 1\right)$

45) b)

$2\cosh^2 x + 3\sinh x - 4 = 0$

$\Rightarrow 2(1 + \sinh^2 x) + 3\sinh x - 4 = 0$

$\Rightarrow 2 + 2\sinh^2 x + 3\sinh x - 4 = 0$

$\Rightarrow 2\sinh^2 x + 3\sinh x - 2 = 0$

$\Rightarrow (2\sinh x - 1)(\sinh x + 2) = 0$

$\Rightarrow \sinh x = \frac{1}{2}$ or $\sinh x = -2$

Consider $\sinh x = \frac{1}{2}$ first:

$\Rightarrow \frac{e^x - e^{-x}}{2} = \frac{1}{2}$

$\Rightarrow e^x - e^{-x} = 1$

$\Rightarrow e^{2x} - 1 = e^x$

$\Rightarrow e^{2x} - e^x - 1 = 0$

$\Rightarrow \left(e^x - \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 - 1 = 0$

$\Rightarrow \left(e^x - \frac{1}{2}\right)^2 = \frac{5}{4}$

$\Rightarrow e^x = \frac{1}{2} \pm \frac{\sqrt{5}}{2}$

$\Rightarrow x = \ln\left(\frac{1}{2} \pm \frac{\sqrt{5}}{2}\right)$

$\Rightarrow x = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$ or $x = \ln\left(\frac{1}{2} - \frac{\sqrt{5}}{2}\right)$

However, $\frac{1}{2} - \frac{\sqrt{5}}{2} < 0$, so the only real solution is $x = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$

Now consider $\sinh x = -2$:

$\Rightarrow \frac{e^x - e^{-x}}{2} = -2$

$\Rightarrow e^x - e^{-x} = -4$

$\Rightarrow e^{2x} - 1 = -4e^x$

$\Rightarrow e^{2x} + 4e^x - 1 = 0$

$\Rightarrow (e^x + 2)^2 - 2^2 - 1 = 0$

$\Rightarrow (e^x + 2)^2 = 5$

$\Rightarrow e^x = -2 \pm \sqrt{5}$

$\Rightarrow x = \ln(-2 + \sqrt{5})$ or $x = \ln(-2 - \sqrt{5})$

However, $-2 - \sqrt{5} < 0$, so the only real solution is $x = \ln(-2 + \sqrt{5})$

So $x = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$ and $x = \ln(-2 + \sqrt{5})$ solve the original equation.

45) c)

$5\sinh x = 2\cosh^2 x$

$\Rightarrow 5\sinh x = 2 \times (1 + \sinh^2 x)$

$\Rightarrow 5\sinh x = 2 + 2\sinh^2 x$

$\Rightarrow 2\sinh^2 x - 5\sinh x + 2 = 0$

$\Rightarrow (2\sinh x - 1)(\sinh x - 2) = 0$

$\Rightarrow \sinh x = \frac{1}{2}$ or $\sinh x = 2$

Consider $\sinh x = \frac{1}{2}$ first:

$\Rightarrow \frac{e^x - e^{-x}}{2} = \frac{1}{2}$

$\Rightarrow e^x - e^{-x} = 1$

$\Rightarrow e^{2x} - 1 = e^x$

$\Rightarrow e^{2x} - e^x - 1 = 0$

$\Rightarrow \left(e^x - \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 - 1 = 0$

$\Rightarrow \left(e^x - \frac{1}{2}\right)^2 = \frac{5}{4}$

$\Rightarrow e^x = \frac{1}{2} \pm \frac{\sqrt{5}}{2}$

$\Rightarrow x = \ln\left(\frac{1}{2} \pm \frac{\sqrt{5}}{2}\right)$

$\Rightarrow x = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$ or $x = \ln\left(\frac{1}{2} - \frac{\sqrt{5}}{2}\right)$

However, $\frac{1}{2} - \frac{\sqrt{5}}{2} < 0$, so the only real solution is $x = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$

Now consider $\sinh x = 2$:

$\Rightarrow \frac{e^x - e^{-x}}{2} = 2$

$\Rightarrow e^x - e^{-x} = 4$

$\Rightarrow e^{2x} - 1 = 4e^x$

$\Rightarrow e^{2x} - 4e^x - 1 = 0$

$\Rightarrow (e^x - 2)^2 - 2^2 - 1 = 0$

$\Rightarrow (e^x - 2)^2 = 5$

$\Rightarrow e^x = 2 \pm \sqrt{5}$

$\Rightarrow x = \ln(2 + \sqrt{5})$ or $x = \ln(2 - \sqrt{5})$

However, $2 - \sqrt{5} < 0$, so the only real solution is $x = \ln(2 + \sqrt{5})$

So $x = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$ and $x = \ln(2 + \sqrt{5})$ solve the original equation.

45) d)

$3\sinh^2 x = 2\cosh x + 5$

$\Rightarrow 3(\cosh^2 x - 1) = 2\cosh x + 5$

$\Rightarrow 3\cosh^2 x - 3 = 2\cosh x + 5$

$\Rightarrow 3\cosh^2 x - 2\cosh x - 8 = 0$

$\Rightarrow (3\cosh x + 4)(\cosh x - 2) = 0$

$\Rightarrow \cosh x = -\frac{4}{3}$ or $\cosh x = 2$

Consider $\cosh x = -\frac{4}{3}$ first:

$\cosh x \geq 1$ for all real values of x, so $\cosh x = -\frac{4}{3}$ has no real solutions.

Now consider $\cosh x = 2$:

$\Rightarrow \frac{e^x - e^{-x}}{2} = 2$

$\Rightarrow e^x - e^{-x} = 4$

$\Rightarrow e^{2x} - 1 = 4e^x$

$\Rightarrow e^{2x} - 4e^x - 1 = 0$

$\Rightarrow (e^x - 2)^2 - 2^2 - 1 = 0$

$\Rightarrow (e^x - 2)^2 = 5$

$\Rightarrow e^x = 2 \pm \sqrt{5}$

$\Rightarrow x = \ln(2 \pm \sqrt{5})$

$\Rightarrow x = \ln(2 + \sqrt{5})$ or $x = \ln(2 - \sqrt{5})$

However, $20 - \sqrt{5} < 0$, so the only real solution is $x = \ln(2 + \sqrt{5})$

46) a)

$\cosh 2x = 3 + 7\cosh x$

$\Rightarrow 2\cosh^2 x - 1 = 3 + 7\cosh x$

$\Rightarrow 2\cosh^2 x - 7\cosh x - 4 = 0$

$\Rightarrow (2\cosh x + 1)(\cosh x - 4) = 0$

$\Rightarrow \cosh x = -\frac{1}{2}$ or $\cosh x = 4$

$\cosh x \geq 1$ for all real values of x, so $\cosh x = -\frac{1}{2}$ has no real solutions.

Now consider $\cosh x = 4$:

$\Rightarrow \frac{e^x + e^{-x}}{2} = 4$

$\Rightarrow e^x + e^{-x} = 8$

$\Rightarrow e^{2x} + 1 = 8e^x$

$\Rightarrow e^{2x} - 8e^x + 1 = 0$

$\Rightarrow (e^x - 4)^2 - 4^2 + 1 = 0$

$\Rightarrow (e^x - 4)^2 = 15$

$\Rightarrow e^x = 4 \pm \sqrt{15}$

$\Rightarrow x = \ln(4 \pm \sqrt{15})$

$\Rightarrow x = \ln(4 + \sqrt{15})$ or $x = \ln(4 - \sqrt{15})$

46) b)

$5\cosh 2x + 8\sinh x = 7$

$\Rightarrow 5(1 + 2\sinh^2 x) + 8\sinh x = 7$

$\Rightarrow 5 + 10\sinh^2 x + 8\sinh x = 7$

$\Rightarrow 10\sinh^2 x + 8\sinh x - 2 = 0$

$\Rightarrow 5\sinh^2 x + 4\sinh x - 1 = 0$

$\Rightarrow (5\sinh x - 1)(\sinh x + 1) = 0$

$\Rightarrow \sinh x = \frac{1}{5}$ or $\sinh x = -1$

First consider $\sinh x = \frac{1}{5}$:

$\Rightarrow \frac{e^x - e^{-x}}{2} = \frac{1}{5}$

$\Rightarrow e^x - e^{-x} = \frac{2}{5}$

$\Rightarrow e^{2x} - 1 = \frac{2}{5}e^x$

$\Rightarrow e^{2x} - \frac{2}{5}e^x - 1 = 0$

$\Rightarrow \left(e^x - \frac{1}{5}\right)^2 - \left(\frac{1}{5}\right)^2 - 1 = 0$

$\Rightarrow \left(e^x - \frac{1}{5}\right)^2 = \frac{26}{25}$

$\Rightarrow e^x = \frac{1}{5} \pm \frac{\sqrt{26}}{5}$

$\Rightarrow x = \ln\left(\frac{1}{5} \pm \frac{\sqrt{26}}{5}\right)$

$\Rightarrow x = \ln\left(\frac{1}{5} + \frac{\sqrt{26}}{5}\right)$ or $x = \ln\left(\frac{1}{5} - \frac{\sqrt{26}}{5}\right)$

However, $\frac{1}{5} - \frac{\sqrt{26}}{5} < 0$, so the only real solution is $x = \ln\left(\frac{1}{5} + \frac{\sqrt{26}}{5}\right)$

Now consider $\sinh x = -1$:

$\Rightarrow \frac{e^x - e^{-x}}{2} = -1$

$\Rightarrow e^x - e^{-x} = -2$

$\Rightarrow e^{2x} - 1 = -2e^x$

$\Rightarrow e^{2x} + 2e^x - 1 = 0$

$\Rightarrow (e^x + 1)^2 - 1^2 - 1 = 0$

$\Rightarrow (e^x + 1)^2 = 2$

$\Rightarrow e^x = -1 \pm \sqrt{2}$

$\Rightarrow x = \ln(-1 \pm \sqrt{2})$

$\Rightarrow x = \ln(-1 + \sqrt{2})$ or $x = \ln(-1 - \sqrt{2})$

However, $-1 - \sqrt{2} < 0$, so the only real solution is $x = \ln(-1 + \sqrt{2})$

So $x = \ln\left(\frac{1}{5} + \frac{\sqrt{26}}{5}\right)$ and $x = \ln(-1 + \sqrt{2})$ solve the original equation.

46) c)

$8\cosh 2x = \cosh x + 9$

$\Rightarrow 8(2\cosh^2 x - 1) = \cosh x + 9$

$\Rightarrow 16\cosh^2 x - 8 = \cosh x + 9$

$\Rightarrow 16\cosh^2 x - \cosh x - 17 = 0$

$\Rightarrow (16\cosh x - 17)(\cosh x + 1) = 0$

$\Rightarrow \cosh x = \frac{17}{16}$ or $\cosh x = -1$

$\cosh x \geq 1$ for all real values of x, so $\cosh x = -1$ has no real solutions.

Now consider $\cosh x = \frac{17}{16}$:

$\Rightarrow \frac{e^x + e^{-x}}{2} = \frac{17}{16}$

$\Rightarrow e^x + e^{-x} = \frac{17}{8}$

$\Rightarrow e^{2x} + 1 = \frac{17}{8}e^x$

$\Rightarrow e^{2x} - \frac{17}{8}e^x + 1 = 0$

$\Rightarrow \left(e^x - \frac{17}{16}\right)^2 - \left(\frac{17}{16}\right)^2 + 1 = 0$

$\Rightarrow \left(e^x - \frac{17}{16}\right)^2 = \frac{33}{256}$

$\Rightarrow e^x = \frac{17}{16} \pm \frac{\sqrt{33}}{16}$

$\Rightarrow x = \ln\left(\frac{17}{16} \pm \frac{\sqrt{33}}{16}\right)$

$\Rightarrow x = \ln\left(\frac{17}{16} + \frac{\sqrt{33}}{16}\right)$ or $x = \ln\left(\frac{17}{16} - \frac{\sqrt{33}}{16}\right)$

46) d)

$\sinh x = 5 - 2\cosh 2x$

$\Rightarrow \sinh x = 5 - 2(1 + 2\sinh^2 x)$

$\Rightarrow \sinh x = 5 - 2 - 4\sinh^2 x$

$\Rightarrow 4\sinh^2 x + \sinh x - 3 = 0$

$\Rightarrow (4\sinh x - 3)(\sinh x + 1) = 0$

$\Rightarrow \sinh x = \frac{3}{4}$ or $\sinh x = -1$

First consider $\sinh x = \frac{3}{4}$:

$\Rightarrow \frac{e^x - e^{-x}}{2} = \frac{3}{4}$

$\Rightarrow e^x - e^{-x} = \frac{3}{2}$

$\Rightarrow e^{2x} - 1 = \frac{3}{2}e^x$

$\Rightarrow e^{2x} - \frac{3}{2}e^x - 1 = 0$

$\Rightarrow \left(e^x - \frac{3}{4}\right)^2 - \left(\frac{3}{4}\right)^2 - 1 = 0$

$\Rightarrow \left(e^x - \frac{3}{4}\right)^2 = \frac{25}{16}$

$\Rightarrow e^x = \frac{3}{4} \pm \frac{5}{4}$

$\Rightarrow x = \ln\left(\frac{3}{4} \pm \frac{5}{4}\right)$

However, $\frac{3}{4} - \frac{5}{4} = -\frac{1}{2} < 0$, so the only real solution is $x = \ln\left(\frac{3}{4} + \frac{5}{4}\right) = \ln 2$

Now consider $\sinh x = -1$:

$\Rightarrow \frac{e^x - e^{-x}}{2} = -1$

$\Rightarrow e^x - e^{-x} = -2$

$\Rightarrow e^{2x} - 1 = -2e^x$

$\Rightarrow e^{2x} + 2e^x - 1 = 0$

$\Rightarrow (e^x + 1)^2 - 1^2 - 1 = 0$

$\Rightarrow (e^x + 1)^2 = 2$

$\Rightarrow e^x = -1 \pm \sqrt{2}$

$\Rightarrow x = \ln(-1 \pm \sqrt{2})$

$\Rightarrow x = \ln(-1 + \sqrt{2})$ or $x = \ln(-1 - \sqrt{2})$

However, $-1 - \sqrt{2} < 0$, so the only real solution is $x = \ln(-1 + \sqrt{2})$

So $x = \ln 2$ and $x = \ln(-1 + \sqrt{2})$ solve the original equation.

47) a)

$\sinh 2x = 20\sinh x$

$\Rightarrow 2\sinh x\cosh x = 20\sinh x$

$\Rightarrow \sinh x\cosh x = 10\sinh x$

$\Rightarrow \sinh x\cosh x - 10\sinh x = 0$

$\Rightarrow \sinh x\,(\cosh x - 10) = 0$

$\Rightarrow \sinh x = 0$ or $\cosh x = 10$

First consider $\sinh x = 0$:

$\Rightarrow \frac{e^x - e^{-x}}{2} = 0$

$\Rightarrow e^x - e^{-x} = 0$

$\Rightarrow e^{2x} - 1 = 0$

$\Rightarrow e^{2x} = 1$

$\Rightarrow 2x = 0$

$\Rightarrow x = 0$

Now consider $\cosh x = 10$:

$\Rightarrow \frac{e^x + e^{-x}}{2} = 10$

$\Rightarrow e^x + e^{-x} = 20$

$\Rightarrow e^{2x} + 1 = 20e^x$

$\Rightarrow e^{2x} - 20e^x + 1 = 0$

$\Rightarrow (e^x - 10)^2 - 10^2 + 1 = 0$

$\Rightarrow (e^x - 10)^2 = 99$

$\Rightarrow e^x = 10 \pm 3\sqrt{11}$

$\Rightarrow x = \ln\left(10 \pm 3\sqrt{11}\right)$

$\Rightarrow x = \ln\left(10 + 3\sqrt{11}\right)$ or $x = \ln\left(10 - 3\sqrt{11}\right)$

So $x = 0$, $x = \ln\left(10 + 3\sqrt{11}\right)$ and $x = \ln\left(10 - 3\sqrt{11}\right)$ solve the original equation.

47) b)

$\frac{2}{3}\sinh 2x = \cosh x$

$\Rightarrow \frac{2}{3} \times 2\sinh x \cosh x = \cosh x$

$\Rightarrow \frac{4}{3}\sinh x \cosh x = \cosh x$

$\Rightarrow \frac{4}{3}\sinh x \cosh x - \cosh x = 0$

$\Rightarrow \cosh x\left(\frac{4}{3}\sinh x - 1\right) = 0$

$\Rightarrow \cosh x = 0$ or $\sinh x = \frac{3}{4}$

$\cosh x \geq 1$ for all real values of x, so $\cosh x = 0$ has no real solutions.

Now consider $\sinh x = \frac{3}{4}$:

$\Rightarrow \frac{e^x - e^{-x}}{2} = \frac{3}{4}$

$\Rightarrow e^x - e^{-x} = \frac{3}{2}$

$\Rightarrow e^{2x} - 1 = \frac{3}{2}e^x$

$\Rightarrow e^{2x} - \frac{3}{2}e^x - 1 = 0$

$\Rightarrow \left(e^x - \frac{3}{4}\right)^2 - \left(\frac{3}{4}\right)^2 - 1 = 0$

$\Rightarrow \left(e^x - \frac{3}{4}\right)^2 = \frac{25}{16}$

$\Rightarrow e^x = \frac{3}{4} \pm \frac{5}{4}$

$\Rightarrow x = \ln\left(\frac{3}{4} \pm \frac{5}{4}\right)$

However, $\frac{3}{4} - \frac{5}{4} = -\frac{1}{2} < 0$, so the only real solution is $x = \ln\left(\frac{3}{4} + \frac{5}{4}\right) = \ln 2$

48) a)

The first mistake is that Dylan substitutes $\cosh^2 x$ for $1 - \sinh^2 x$, when it should be substituted for $1 + \sinh^2 x$. The second mistake is on the 3rd line, where $+4$ has become -4 when attempting to complete the square.

48) b)

$3\sinh x + \cosh^2 x = 5$

$\Rightarrow 3\sinh x + 1 + \sinh^2 x = 5$

$\Rightarrow \sinh^2 x + 3\sinh x - 4 = 0$

$\Rightarrow (\sinh x + 4)(\sinh x - 1) = 0$

First consider $\sinh x = -4$:

$\Rightarrow \frac{e^x - e^{-x}}{2} = -4$

$\Rightarrow e^x - e^{-x} = -8$

$\Rightarrow e^{2x} - 1 = -8e^x$

$\Rightarrow e^{2x} + 8e^x - 1 = 0$

$\Rightarrow e^x = -4 \pm \sqrt{17}$

$\Rightarrow x = \ln(-4 \pm \sqrt{17})$

However, $-4 - \sqrt{17} < 0$, so the only real solution is

$x = \ln(-4 + \sqrt{17})$

Now consider $\sinh x = 1$:

$\Rightarrow \frac{e^x - e^{-x}}{2} = 1$

$\Rightarrow e^x - e^{-x} = 2$

$\Rightarrow e^{2x} - 1 = 2e^x$

$\Rightarrow e^{2x} - 2e^x - 1 = 0$

$\Rightarrow e^x = 1 \pm \sqrt{2}$

$\Rightarrow x = \ln(1 \pm \sqrt{2})$

However, $1 - \sqrt{2} < 0$, so the only real solution is

$x = \ln(1 + \sqrt{2})$

So $x = \ln(-4 + \sqrt{17})$ and $x = \ln(1 + \sqrt{2})$ solve the original equation.

49) a)

$\cosh 2x \equiv 2\cosh^2 x - 1 \Rightarrow \cosh^2 x \equiv \frac{\cosh 2x + 1}{2}$

$\therefore \cosh^2 2x \equiv \frac{\cosh 4x + 1}{2}$

LHS: $8\cosh^4 x \equiv 8(\cosh^2 x)^2 \equiv 8\left(\frac{\cosh 2x + 1}{2}\right)^2$

$\equiv 8 \times \frac{(\cosh 2x + 1)^2}{4} \equiv 2(\cosh 2x + 1)^2$

$\equiv 2(\cosh^2 2x + 2\cosh 2x + 1)$

$\equiv 2\left(\frac{\cosh 4x + 1}{2} + 2\cosh 2x + 1\right)$

$\equiv \cosh 4x + 1 + 4\cosh 2x + 2$

$\equiv \cosh 4x + 4\cosh 2x + 3$: RHS

49) b)

$8\cosh^4 x \equiv \cosh 4x + 4\cosh 2x + 3$

$\Rightarrow \cosh 4x \equiv 8\cosh^4 x - 4\cosh 2x - 3$

$\Rightarrow \cosh 4x \equiv 8\cosh^4 x - 4(2\cosh^2 x - 1) - 3$

$\Rightarrow \cosh 4x \equiv 8\cosh^4 x - 8\cosh^2 x + 1$

Then:

$2\cosh 4x - 23\cosh 2x - 33 = 0$

$2(8\cosh^4 x - 8\cosh^2 x + 1) - 23(2\cosh^2 x - 1) - 33 = 0$

$\Rightarrow 16\cosh^4 x - 16\cosh^2 x + 2 - 46\cosh^2 x + 23 - 33 = 0$

$\Rightarrow 16\cosh^4 x - 62\cosh^2 x - 8 = 0$

$\Rightarrow 8\cosh^4 x - 31\cosh^2 x - 4 = 0$

$\Rightarrow (8\cosh^2 x + 1)(\cosh^2 x - 4) = 0$

$\Rightarrow \cosh^2 x = -\frac{1}{8}$ or $\cosh^2 x = 4$

First consider $\cosh^2 x = -\frac{1}{8}$:

$\cosh^2 x \geq 1$ for all real values of x, so $\cosh^2 x = -\frac{1}{8}$ has no real solutions.

Now consider $\cosh^2 x = 4 \Rightarrow \cosh x = \pm 2$:

Firstly, $\cosh x = -2$:

$\cosh x \geq 1$ for all real values of x, so $\cosh x = -2$ has no real solutions.

Secondly, $\cosh x = 2$:

$\Rightarrow \frac{e^x + e^{-x}}{2} = 2$

$\Rightarrow e^x + e^{-x} = 4$

$\Rightarrow e^{2x} + 1 = 4e^x$

$\Rightarrow e^{2x} - 4e^x + 1 = 0$

$\Rightarrow (e^x - 2)^2 - 2^2 + 1 = 0$

$\Rightarrow (e^x - 2)^2 = 3$

$\Rightarrow e^x = 2 \pm \sqrt{3}$

$\Rightarrow x = \ln(2 + \sqrt{3})$ or $x = \ln(2 - \sqrt{3})$

50) a)

$\sinh x\,(\sinh x + \cosh x) = 4$

$\Rightarrow \frac{e^x - e^{-x}}{2} \times \left(\frac{e^x - e^{-x}}{2} + \frac{e^x + e^{-x}}{2}\right) = 4$

$\Rightarrow \frac{e^x - e^{-x}}{2} \times \left(\frac{e^x - e^{-x} + e^x + e^{-x}}{2}\right) = 4$

$\Rightarrow \frac{e^x - e^{-x}}{2} \times \left(\frac{2e^x}{2}\right) = 4$

$\Rightarrow \frac{e^x - e^{-x}}{2} \times e^x = 4$

$\Rightarrow (e^x - e^{-x})e^x = 8$

$\Rightarrow e^{2x} - 1 = 8$

$\Rightarrow e^{2x} = 9$

$\Rightarrow 2x = \ln 9$

$\Rightarrow x = \frac{1}{2}\ln 9 = \ln 3$

50) b)

$\cosh x\,(\sinh x + \cosh x) = 13$

$\Rightarrow \frac{e^x + e^{-x}}{2} \times \left(\frac{e^x - e^{-x}}{2} + \frac{e^x + e^{-x}}{2}\right) = 13$

$\Rightarrow \frac{e^x + e^{-x}}{2} \times \left(\frac{e^x - e^{-x} + e^x + e^{-x}}{2}\right) = 13$

$\Rightarrow \frac{e^x + e^{-x}}{2} \times \left(\frac{2e^x}{2}\right) = 13$

$\Rightarrow \frac{e^x + e^{-x}}{2} \times e^x = 13$

$\Rightarrow (e^x + e^{-x})e^x = 26$

$\Rightarrow e^{2x} + 1 = 26$

$\Rightarrow e^{2x} = 25$

$\Rightarrow 2x = \ln 25$

$\Rightarrow x = \frac{1}{2}\ln 25 = \ln 5$

50) c)

$\sinh x\,(5\sinh x - 4\cosh x) = 3$

$\Rightarrow \frac{e^x - e^{-x}}{2} \times \left(5 \times \frac{e^x - e^{-x}}{2} - 4 \times \frac{e^x + e^{-x}}{2}\right) = 3$

$\Rightarrow \frac{e^x - e^{-x}}{2} \times \left(\frac{5(e^x - e^{-x}) - 4(e^x + e^{-x})}{2}\right) = 3$

$\Rightarrow \frac{e^x - e^{-x}}{2} \times \left(\frac{5e^x - 5e^{-x} - 4e^x - 4e^{-x}}{2}\right) = 3$

$\Rightarrow \frac{e^x - e^{-x}}{2} \times \left(\frac{e^x - 9e^{-x}}{2}\right) = 3$

$\Rightarrow (e^x - e^{-x})(e^x - 9e^{-x}) = 12$

$\Rightarrow e^{2x} - 10 + 9e^{-2x} = 12$

$\Rightarrow e^{4x} - 10e^{2x} + 9 = 12e^{2x}$

$\Rightarrow e^{4x} - 22e^{2x} + 9 = 0$

$\Rightarrow (e^{2x} - 11)^2 - 11^2 + 9 = 0$

$\Rightarrow (e^{2x} - 11)^2 = 112$

$\Rightarrow e^{2x} = 11 \pm 4\sqrt{7}$

$\Rightarrow 2x = \ln(11 \pm 4\sqrt{7})$

$\Rightarrow x = \frac{1}{2}\ln(11 \pm 4\sqrt{7})$

$\Rightarrow x = \frac{1}{2}\ln(11 + 4\sqrt{7})$ or $x = \frac{1}{2}\ln(11 - 4\sqrt{7})$

51) a)

$3\tanh 2x - 4\tanh x = 0$

$\Rightarrow 3 \times \frac{2\tanh x}{1+\tanh^2 x} - 4\tanh x = 0$

$\Rightarrow 6\tanh x - 4\tanh x\,(1 + \tanh^2 x) = 0$

$\Rightarrow 6\tanh x - 4\tanh x - 4\tanh^3 x = 0$

$\Rightarrow 2\tanh x - 4\tanh^3 x = 0$

$\Rightarrow \tanh x - 2\tanh^3 x = 0$

$\Rightarrow \tanh x\,(1 - 2\tanh^2 x) = 0$

$\Rightarrow \tanh x = 0$ or $\tanh^2 x = \frac{1}{2}$

First consider $\tanh x = 0$:

$\Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = 0$

$\Rightarrow e^{2x} - 1 = 0$

$\Rightarrow e^{2x} = 1$

$\Rightarrow 2x = 0$

$\Rightarrow x = 0$

Now consider $\tanh^2 x = \frac{1}{2} \Rightarrow \tanh x = \pm\frac{\sqrt{2}}{2}$

Firstly $\tanh x = \frac{\sqrt{2}}{2}$:

$\Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = \frac{\sqrt{2}}{2}$

$\Rightarrow 2(e^{2x} - 1) = \sqrt{2}(e^{2x} + 1)$

$\Rightarrow 2e^{2x} - 2 = \sqrt{2}e^{2x} + \sqrt{2}$

$\Rightarrow \left(2 - \sqrt{2}\right)e^{2x} = 2 + \sqrt{2}$

$\Rightarrow e^{2x} = \frac{2+\sqrt{2}}{2-\sqrt{2}} = 3 + 2\sqrt{2}$

$\Rightarrow 2x = \ln\left(3 + 2\sqrt{2}\right)$

$\Rightarrow x = \frac{1}{2}\ln\left(3 + 2\sqrt{2}\right)$

Secondly $\tanh x = -\frac{\sqrt{2}}{2}$:

$\Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = -\frac{\sqrt{2}}{2}$

$\Rightarrow 2(e^{2x} - 1) = -\sqrt{2}(e^{2x} + 1)$

$\Rightarrow 2e^{2x} - 2 = -\sqrt{2}e^{2x} - \sqrt{2}$

$\Rightarrow \left(2 + \sqrt{2}\right)e^{2x} = 2 - \sqrt{2}$

$\Rightarrow e^{2x} = \frac{2-\sqrt{2}}{2+\sqrt{2}} = 3 - 2\sqrt{2}$

$\Rightarrow 2x = \ln\left(3 - 2\sqrt{2}\right)$

$\Rightarrow x = \frac{1}{2}\ln\left(3 - 2\sqrt{2}\right)$

So $x = \frac{1}{2}\ln\left(3 + 2\sqrt{2}\right)$ and $x = \frac{1}{2}\ln\left(3 - 2\sqrt{2}\right)$ solve the original equation.

51) b)

$10\tanh 2x = 9 + \tanh x$

$\Rightarrow 10 \times \frac{2\tanh x}{1+\tanh^2 x} = 9 + \tanh x$

$\Rightarrow 20\tanh x = (9 + \tanh x)(1 + \tanh^2 x)$

$\Rightarrow 20\tanh x = 9 + 9\tanh^2 x + \tanh x + \tanh^3 x$

$\Rightarrow \tanh^3 x + 9\tanh^2 x - 19\tanh x + 9 = 0$

$\Rightarrow (\tanh x - 1)(\tanh^2 x + 10\tanh x - 9) = 0$

$\Rightarrow \tanh x = 1$ or $\tanh^2 x + 10\tanh x - 9 = 0$

First consider $\tanh x = 1$:

$-1 < \tanh x < 1$ for all real values of x, so $\tanh x = 1$ has no real solutions.

Now consider $\tanh^2 x + 10\tanh x - 9 = 0$

$\Rightarrow (\tanh x + 5)^2 - 5^2 - 9 = 0$

$\Rightarrow (\tanh x + 5)^2 = 34$

$\Rightarrow \tanh x = -5 \pm \sqrt{34}$

$\Rightarrow \tanh x = -5 + \sqrt{34}$ or $\tanh x = -5 - \sqrt{34}$

Firstly, $\tanh x = -5 - \sqrt{34}$:

$-1 < \tanh x < 1$ for all real values of x, so $\tanh x = -5 - \sqrt{34}$ has no real solutions.

Secondly, $\tanh x = -5 + \sqrt{34}$:

$\Rightarrow \frac{e^{2x}-1}{e^{2x}+1} = -5 + \sqrt{34}$

$\Rightarrow e^{2x} - 1 = \left(-5 + \sqrt{34}\right)(e^{2x} + 1)$

$\Rightarrow e^{2x} - 1 = \left(-5 + \sqrt{34}\right)e^{2x} + \left(-5 + \sqrt{34}\right)$

$\Rightarrow \left(1 - \left(-5 + \sqrt{34}\right)\right)e^{2x} = -4 + \sqrt{34}$

$\Rightarrow \left(6 - \sqrt{34}\right)e^{2x} = -4 + \sqrt{34}$

$\Rightarrow e^{2x} = \frac{-4+\sqrt{34}}{6-\sqrt{34}} = 5 + \sqrt{34}$

$\Rightarrow 2x = \ln\left(5 + \sqrt{34}\right)$

$\Rightarrow x = \frac{1}{2}\ln\left(5 + \sqrt{34}\right)$

52)

LHS: $\sinh x \cosh y + \cosh x \sinh y$

$$\equiv \frac{e^x - e^{-x}}{2} \times \frac{e^y + e^{-y}}{2} + \frac{e^x + e^{-x}}{2} \times \frac{e^y - e^{-y}}{2}$$

$$\equiv \frac{(e^x - e^{-x})(e^y + e^{-y}) + (e^x + e^{-x})(e^y - e^{-y})}{4}$$

$$\equiv \frac{e^x e^y + e^x e^{-y} - e^{-x} e^y - e^{-x} e^{-y} + e^x e^y - e^x e^{-y} + e^{-x} e^y - e^{-x} e^{-y}}{4}$$

$$\equiv \frac{2e^x e^y - 2e^{-x} e^{-y}}{4}$$

$$\equiv \frac{e^x e^y - e^{-x} e^{-y}}{2}$$

$$\equiv \frac{e^{x+y} - e^{-x-y}}{2}$$

$$\equiv \frac{e^{x+y} - e^{-(x+y)}}{2} \equiv \sinh(x + y) \text{ :RHS}$$

53)

LHS: $\cosh x \cosh y - \sinh x \sinh y$

$$\equiv \frac{e^x + e^{-x}}{2} \times \frac{e^y + e^{-y}}{2} - \frac{e^x - e^{-x}}{2} \times \frac{e^y - e^{-y}}{2}$$

$$\equiv \frac{(e^x + e^{-x})(e^y + e^{-y}) - (e^x - e^{-x})(e^y - e^{-y})}{4}$$

$$\equiv \frac{e^x e^y + e^x e^{-y} + e^{-x} e^y + e^{-x} e^{-y} - (e^x e^y - e^x e^{-y} - e^{-x} e^y + e^{-x} e^{-y})}{4}$$

$$\equiv \frac{e^x e^y + e^x e^{-y} + e^{-x} e^y + e^{-x} e^{-y} - e^x e^y + e^x e^{-y} + e^{-x} e^y - e^{-x} e^{-y}}{4}$$

$$\equiv \frac{2e^x e^{-y} + 2e^{-x} e^y}{4}$$

$$\equiv \frac{e^x e^{-y} + e^{-x} e^y}{2}$$

$$\equiv \frac{e^{x-y} + e^{-x+y}}{2}$$

$$\equiv \frac{e^{x-y} + e^{-(x-y)}}{2} \equiv \cosh(x - y) \text{ :RHS}$$

54) a)

Using the compound angle formulae proved in Q48:

$\sinh(x + \ln 5) = \cosh x$

$\Rightarrow \sinh x \cosh(\ln 5) + \cosh x \sinh(\ln 5) = \cosh x$

$\Rightarrow \sinh x \times \frac{e^{\ln 5} + e^{-\ln 5}}{2} + \cosh x \times \frac{e^{\ln 5} - e^{-\ln 5}}{2} = \cosh x$

$\Rightarrow \sinh x \times \frac{5 + \frac{1}{5}}{2} + \cosh x \times \frac{5 - \frac{1}{5}}{2} = \cosh x$

$\Rightarrow \frac{13}{5}\sinh x + \frac{12}{5}\cosh x = \cosh x$

$\Rightarrow \frac{13}{5}\sinh x + \frac{7}{5}\cosh x = 0$

$\Rightarrow 13\sinh x + 7\cosh x = 0$

$\Rightarrow 13\sinh x = -7\cosh x$

$\Rightarrow \frac{\sinh x}{\cosh x} = -\frac{7}{13}$

$\Rightarrow \tanh x = -\frac{7}{13}$ as required

54) b)

$\frac{e^{2x} - 1}{e^{2x} + 1} = -\frac{7}{13}$

$\Rightarrow 13(e^{2x} - 1) = -7(e^{2x} + 1)$

$\Rightarrow 13e^{2x} - 13 = -7e^{2x} - 7$

$\Rightarrow 20e^{2x} = 6$

$\Rightarrow e^{2x} = \frac{3}{10}$

$\Rightarrow 2x = \ln\frac{3}{10}$

$\Rightarrow x = \frac{1}{2}\ln\frac{3}{10} = \frac{1}{2}\ln\left(\frac{3}{2\times 5}\right) = \frac{1}{2}(\ln 3 - \ln(2 \times 5))$

$= \frac{1}{2}\big(\ln 3 - (\ln 2 + \ln 5)\big) = \frac{1}{3}(\ln 3 - \ln 2 - \ln 5)$

55)

LHS: $\sinh 3x \equiv \sinh(2x + x) \equiv \sinh 2x \cosh x + \cosh 2x \sinh x$

$\equiv (2 \sinh x \cosh x) \cosh x + (1 + 2 \sinh^2 x) \sinh x$

$\equiv 2 \sinh x \cosh^2 x + \sinh x + 2 \sinh^3 x$

$\equiv 2 \sinh x (1 + \sinh^2 x) + \sinh x + 2 \sinh^3 x$

$\equiv 2 \sinh x + 2 \sinh^3 x + \sinh x + 2 \sinh^3 x$

$\equiv 4 \sinh^3 x + 3 \sinh x$:RHS

56)

LHS: $\cosh 3x \equiv \cosh(2x + x) \equiv \cosh 2x \cosh x + \sinh 2x \sinh x$

$\equiv (2 \cosh^2 x - 1) \cosh x + (2 \sinh x \cosh x) \sinh x$

$\equiv 2 \cosh^3 x - \cosh x + 2 \sinh^2 x \cosh x$

$\equiv 2 \cosh^3 x - \cosh x + 2(\cosh^2 x - 1) \cosh x$

$\equiv 2 \cosh^3 x - \cosh x + 2 \cosh^3 x - 2 \cosh x$

$\equiv 4 \cosh^3 x - 3 \cosh x$:RHS

57) a)

$\sinh 3x = 15\sinh x \Rightarrow 4\sinh^3 x + 3\sinh x = 15\sinh x$

$\Rightarrow 4\sinh^3 x - 12\sinh x = 0$

$\Rightarrow \sinh^3 x - 3\sinh x = 0$

$\Rightarrow \sinh x\,(\sinh^2 x - 3) = 0$

$\Rightarrow \sinh x = 0$ or $\sinh x = \pm\sqrt{3}$

If $\sinh x = 0 \Rightarrow x = 0$

If $\sinh x = \sqrt{3} \Rightarrow \frac{e^x - e^{-x}}{2} = \sqrt{3}$

$\Rightarrow e^x - e^{-x} = 2\sqrt{3}$

$\Rightarrow e^{2x} - 1 = 2\sqrt{3}e^x$

$\Rightarrow e^{2x} - 2\sqrt{3}e^x - 1 = 0$

$\Rightarrow \left(e^x - \sqrt{3}\right)^2 - 3 - 1 = 0$

$\Rightarrow \left(e^x - \sqrt{3}\right)^2 = 4$

$\Rightarrow e^x = \sqrt{3} \pm 2$

$\Rightarrow x = \ln\left(\sqrt{3} + 2\right)$ only as $\sqrt{3} - 2 < 0$

If $\sinh x = -\sqrt{3} \Rightarrow \frac{e^x - e^{-x}}{2} = -\sqrt{3}$

$\Rightarrow e^x - e^{-x} = -2\sqrt{3}$

$\Rightarrow e^{2x} - 1 = -2\sqrt{3}e^x$

$\Rightarrow e^{2x} + 2\sqrt{3}e^x - 1 = 0$

$\Rightarrow \left(e^x + \sqrt{3}\right)^2 - 3 - 1 = 0$

$\Rightarrow \left(e^x + \sqrt{3}\right)^2 = 4$

$\Rightarrow e^x = -\sqrt{3} \pm 2$

$\Rightarrow x = \ln\left(-\sqrt{3} + 2\right)$ only as $-\sqrt{3} - 2 < 0$

$\therefore x = 0$ or $x = \ln\left(\sqrt{3} + 2\right)$ or $x = \ln\left(-\sqrt{3} + 2\right)$

57) b)

$\cosh 3x = 5\cosh x \Rightarrow 4\cosh^3 x - 3\cosh x = 5\cosh x$

$\Rightarrow 4\cosh^3 x - 8\cosh x = 0$

$\Rightarrow \cosh^3 x - 2\cosh x = 0$

$\Rightarrow \cosh x\,(\cosh^2 x - 2) = 0$

$\Rightarrow \cosh x = 0$ or $\cosh x = \pm\sqrt{2}$

$\cosh x = 0$ has no real solutions as $\cosh x \geq 1$

If $\cosh x = \sqrt{2} \Rightarrow \frac{e^x + e^{-x}}{2} = \sqrt{2}$

$\Rightarrow e^x + e^{-x} = 2\sqrt{2}$

$\Rightarrow e^{2x} + 1 = 2\sqrt{2}e^x$

$\Rightarrow e^{2x} - 2\sqrt{2}e^x + 1 = 0$

$\Rightarrow \left(e^x - \sqrt{2}\right)^2 - 2 + 1 = 0$

$\Rightarrow \left(e^x - \sqrt{2}\right)^2 = 1$

$\Rightarrow e^x = \sqrt{2} \pm 1$

$\Rightarrow x = \ln\left(\sqrt{2}+1\right)$ or $\ln\left(\sqrt{2}-1\right)$

If $\cosh x = -\sqrt{2}$ has no real solutions as $\cosh x \geq 1$

$\therefore x = \ln\left(\sqrt{2}+1\right)$ or $\ln\left(\sqrt{2}-1\right)$

58) a)

$\sinh x = 3$

$\cosh^2 x - \sinh^2 x = 1$

$\Rightarrow \cosh^2 x - 3^2 = 1$

$\Rightarrow \cosh^2 x = 10$

$\Rightarrow \cosh x = \sqrt{10}$ as $\cosh x \neq -\sqrt{10}$

58) b)

$\tanh x = \frac{\sinh x}{\cosh x}$

$\Rightarrow \tanh x = \frac{3}{\sqrt{10}} = \frac{3\sqrt{10}}{10}$

59) a)

$\cosh x = 5$

$\cosh^2 x - \sinh^2 x = 1$

$\Rightarrow 5^2 - \sinh^2 x = 1$

$\Rightarrow \sinh^2 x = 24$

$\Rightarrow \sinh x = \sqrt{24} = 2\sqrt{6}$

$\sinh x \neq -\sqrt{24}$ when $x > 0$

59) b)

$\tanh x = \frac{\sinh x}{\cosh x}$

$\Rightarrow \tanh x = \frac{2\sqrt{6}}{5}$

60) a)

$\tanh x = \frac{1}{3}$

$\frac{\sinh x}{\cosh x} = \frac{1}{3} \Rightarrow 3\sinh x = \cosh x$ (1)

$\cosh^2 x - \sinh^2 x = 1$

$\therefore (3\sinh x)^2 - \sinh^2 x = 1$

$\Rightarrow 9\sinh^2 x - \sinh^2 x = 1$

$\Rightarrow 8\sinh^2 x = 1 \Rightarrow \sinh^2 x = \frac{1}{8}$

$\Rightarrow \sinh x = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}$ only as $x > 0$

60) b)

Substitute into (1)

$3 \times \frac{\sqrt{2}}{4} = \cosh x \Rightarrow \cosh x = \frac{3\sqrt{2}}{4}$

61)

$2\cosh x + \sinh x = 2$ (1)

$4\cosh^2 x - 11\sinh x = 4$ (2)

(1) $2\cosh x + \sinh x = 2 \Rightarrow \sinh x = 2 - 2\cosh x$

Substitute into (2):

$4\cosh^2 x - 11(2 - 2\cosh x) = 4$

$\Rightarrow 4\cosh^2 x - 22 + 22\cosh x = 4$

$\Rightarrow 4\cosh^2 x + 22\cosh x - 26 = 0$

$\Rightarrow 2\cosh^2 x + 11\cosh x - 13 = 0$

$\Rightarrow (2\cosh x + 13)(\cosh x - 1) = 0$

$\Rightarrow \cosh x = -\frac{13}{2}$ or $\cosh x = 1$

$\cosh x \neq -\frac{13}{2}$ as $\cosh x \geq 1$

If $\cosh x = 1 \Rightarrow \frac{e^x + e^{-x}}{2} = 1$

$\Rightarrow e^x + e^{-x} = 2$

$\Rightarrow e^{2x} + 1 = 2e^x$

$\Rightarrow e^{2x} - 2e^x + 1 = 0$

$\Rightarrow (e^x - 1)^2 = 0$

$\Rightarrow e^x = 1 \Rightarrow x = 0$

Check in (1):

$2\cosh(0) + \sinh(0) = 2 \times 1 + 0 = 2$ as required

$\therefore x = 0$

62)

$2\cosh^2 x - \sinh x = 8$ (1)

$2\cosh^2 x - 3\sinh x = 4$ (2)

Rearrange (1):

$\cosh^2 x = \frac{8+\sinh x}{2}$

Substitute into (2):

$2\left(\frac{8+\sinh x}{2}\right) - 3\sinh x = 4$

$\Rightarrow 8 + \sinh x - 3\sinh x = 4$

$\Rightarrow -2\sinh x = -4$

$\Rightarrow \sinh x = 2$

$\Rightarrow \frac{e^x - e^{-x}}{2} = 2$

$\Rightarrow e^x - e^{-x} = 4$

$\Rightarrow e^{2x} - 1 = 4e^x$

$\Rightarrow e^{2x} - 4e^x - 1 = 0$

$\Rightarrow (e^x - 2)^2 - 4 - 1 = 0$

$\Rightarrow (e^x - 2)^2 = 5$

$\Rightarrow e^x = 2 \pm \sqrt{5}$

$e^x \neq 2 - \sqrt{5}$ as $2 - \sqrt{5} < 0$ and $e^x > 0$

$\therefore x = \ln(2 + \sqrt{5})$

Check in (1):

$2\cosh^2(\ln(2 + \sqrt{5})) - \sinh(\ln(2 + \sqrt{5}))$

$= 2\left(\frac{e^{\ln(2+\sqrt{5})} + e^{-\ln(2+\sqrt{5})}}{2}\right)^2 - \frac{e^{\ln(2+\sqrt{5})} - e^{-\ln(2+\sqrt{5})}}{2}$

$= 2\left(\frac{e^{\ln(2+\sqrt{5})} + e^{\ln(2+\sqrt{5})^{-1}}}{2}\right)^2 - \frac{e^{\ln(2+\sqrt{5})} - e^{\ln(2+\sqrt{5})^{-1}}}{2}$

$= 2\left(\frac{(2+\sqrt{5}) + (-2+\sqrt{5})}{2}\right)^2 - \frac{(2+\sqrt{5}) - (-2+\sqrt{5})}{2}$

$= 2(\sqrt{5})^2 - \frac{4}{2} = 10 - 2 = 8$ as required

Check in (2):

$2\cosh^2(\ln(2 + \sqrt{5})) - 3\sinh(\ln(2 + \sqrt{5}))$

$= 2\left(\frac{e^{\ln(2+\sqrt{5})} + e^{-\ln(2+\sqrt{5})}}{2}\right)^2 - 3 \times \frac{e^{\ln(2+\sqrt{5})} - e^{-\ln(2+\sqrt{5})}}{2}$

$= 2\left(\frac{e^{\ln(2+\sqrt{5})}+e^{\ln(2+\sqrt{5})^{-1}}}{2}\right)^2 - 3 \times \frac{e^{\ln(2+\sqrt{5})}-e^{\ln(2+\sqrt{5})^{-1}}}{2}$

$= 2\left(\frac{(2+\sqrt{5})+(-2+\sqrt{5})}{2}\right)^2 - 3 \times \frac{(2+\sqrt{5})-(-2+\sqrt{5})}{2}$

$= 2\left(\sqrt{5}\right)^2 - 3 \times \frac{4}{2} = 10 - 6 = 4$ as required

$\therefore x = \ln\left(2+\sqrt{5}\right)$

63) a)

LHS: $(\sinh x + \cosh x)^2 \equiv \left(\frac{e^x-e^{-x}}{2} + \frac{e^x+e^{-x}}{2}\right)^2$

$\equiv \left(\frac{e^x-e^{-x}+e^x+e^{-x}}{2}\right)^2 \equiv \left(\frac{2e^x}{2}\right)^2 \equiv (e^x)^2 \equiv e^{2x}$

$\equiv \frac{2e^{2x}}{2} \equiv \frac{e^{2x}-e^{-2x}+e^{2x}+e^{-2x}}{2} \equiv \frac{e^{2x}-e^{-2x}}{2} + \frac{e^{2x}+e^{-2x}}{2} \equiv \sinh 2x + \cosh 2x$: RHS

63) b)

LHS: $(\sinh 2x - \cosh 2x)^2 \equiv \left(\frac{e^{2x}-e^{-2x}}{2} - \frac{e^{2x}+e^{-2x}}{2}\right)^2$

$\equiv \left(\frac{e^{2x}-e^{-2x}-e^{2x}-e^{-2x}}{2}\right)^2$

$\equiv \left(\frac{-2e^{-2x}}{2}\right)^2 \equiv (-e^{-2x})^2 \equiv e^{-4x} \equiv \frac{2e^{-4x}}{2}$

$\equiv \frac{e^{4x}+e^{-4x}-e^{4x}+e^{-4x}}{2}$

$\equiv \frac{e^{4x}+e^{-4x}}{2} - \frac{e^{4x}-e^{-4x}}{2} \equiv \cosh 4x - \sinh 4x$:RHS

63) c)

LHS: $1 - \tanh^2 x \equiv 1 - \frac{\sinh^2 x}{\cosh^2 x}$

$\equiv \frac{\cosh^2 x}{\cosh^2 x} - \frac{\sinh^2 x}{\cosh^2 x} \equiv \frac{\cosh^2 x-\sinh^2 x}{\cosh^2 x} \equiv \frac{1}{\cosh^2 x}$:RHS

63) d)

LHS: $\frac{1}{\tanh^2 x} - 1 \equiv \frac{1}{\frac{\sinh^2 x}{\cosh^2 x}} - 1 \equiv \frac{\cosh^2 x}{\sinh^2 x} - 1$

$\equiv \frac{\cosh^2 x}{\sinh^2 x} - \frac{\sinh^2 x}{\sinh^2 x} \equiv \frac{\cosh^2 x-\sinh^2 x}{\sinh^2 x} \equiv \frac{1}{\sinh^2 x}$:RHS

64)

LHS: $\sinh^3 x \equiv \left(\frac{e^x - e^{-x}}{2}\right)^3$

$\equiv \frac{(e^x - e^{-x})^3}{8}$

$\equiv \frac{e^{3x} - 3e^x + 3e^{-x} - e^{-3x}}{8}$

$\equiv \frac{e^{3x} - e^{-3x}}{8} + \frac{-3e^x + 3e^{-x}}{8}$

$\equiv \frac{1}{4} \times \frac{e^{3x} - e^{-3x}}{2} - \frac{3}{4} \times \frac{e^x - e^{-x}}{2}$

$\equiv \frac{1}{4}\sinh 3x - \frac{3}{4}\sinh x$:RHS

65)

LHS: $\cosh 3x - 3\cosh x \equiv \frac{e^{3x} + e^{-3x}}{2} - 3 \times \frac{e^x + e^{-x}}{2}$

$\equiv \frac{(e^x + e^{-x})(e^{2x} + e^{-2x} - 1)}{2} - 3 \times \frac{e^x + e^{-x}}{2}$

$\equiv \frac{e^x + e^{-x}}{2} \times (e^{2x} + e^{-2x} - 1) - 3 \times \frac{e^x + e^{-x}}{2}$

$\equiv \frac{e^x + e^{-x}}{2}(e^{2x} + e^{-2x} - 1 - 3)$

$\equiv \frac{e^x + e^{-x}}{2}(e^{2x} + e^{-2x} - 4)$

$\equiv \frac{e^x + e^{-x}}{2}\left(2\left(\frac{e^{2x} + e^{-2x}}{2} - 2\right)\right)$

$\equiv 2 \times \frac{e^x + e^{-x}}{2}\left(\frac{e^{2x} + e^{-2x}}{2} - 2\right)$

$\equiv 2\cosh x\,(\cosh 2x - 2)$:RHS

66)

LHS: $\cosh 2x - 2\sinh^2 x$

$\equiv \frac{e^{2x} + e^{-2x}}{2} - 2\left(\frac{e^x - e^{-x}}{2}\right)^2$

$\equiv \frac{e^{2x} + e^{-2x}}{2} - 2 \times \frac{e^{2x} - 2 + e^{-2x}}{4}$

$\equiv \frac{e^{2x} + e^{-2x}}{2} - \frac{e^{2x} + e^{-2x} - 2}{2}$

$\equiv \frac{e^{2x} + e^{-2x}}{2} - \frac{e^{2x} + e^{-2x}}{2} + 1 \equiv 1$:RHS

67)

RHS: $\sinh\left(\frac{x}{2}\right)\left(6\cosh\left(\frac{x}{2}\right)-7\right)$

$\equiv \frac{e^{\frac{x}{2}}-e^{-\frac{x}{2}}}{2}\times\left(6\times\frac{e^{\frac{x}{2}}+e^{-\frac{x}{2}}}{2}-7\right)$

$\equiv \frac{e^{\frac{x}{2}}-e^{-\frac{x}{2}}}{2}\times 6\times\frac{e^{\frac{x}{2}}+e^{-\frac{x}{2}}}{2}-7\times\frac{e^{\frac{x}{2}}-e^{-\frac{x}{2}}}{2}$

$\equiv \frac{3}{2}\times\left(e^{\frac{x}{2}}-e^{-\frac{x}{2}}\right)\left(e^{\frac{x}{2}}+e^{-\frac{x}{2}}\right)-7\times\frac{e^{\frac{x}{2}}-e^{-\frac{x}{2}}}{2}$

$\equiv \frac{3}{2}\times(e^{x}-e^{-x})-7\times\frac{e^{\frac{x}{2}}-e^{-\frac{x}{2}}}{2}$

$\equiv 3\times\frac{e^{x}-e^{-x}}{2}-7\times\frac{e^{\frac{x}{2}}-e^{-\frac{x}{2}}}{2}$

$\equiv 3\sinh x-7\sinh\left(\frac{x}{2}\right)$:LHS

68)

RHS: $4\sinh\left(\frac{x}{4}\right)\cosh\left(\frac{x}{4}\right)\cosh\left(\frac{x}{2}\right)$

$\equiv 4\left(\frac{e^{\frac{x}{4}}-e^{-\frac{x}{4}}}{2}\right)\left(\frac{e^{\frac{x}{4}}+e^{-\frac{x}{4}}}{2}\right)\left(\frac{e^{\frac{x}{2}}+e^{-\frac{x}{2}}}{2}\right)$

$\equiv \frac{1}{2}\left(e^{\frac{x}{4}}-e^{-\frac{x}{4}}\right)\left(e^{\frac{x}{4}}+e^{-\frac{x}{4}}\right)\left(e^{\frac{x}{2}}+e^{-\frac{x}{2}}\right)$

$\equiv \frac{1}{2}\left(e^{\frac{x}{2}}-e^{-\frac{x}{2}}\right)\left(e^{\frac{x}{2}}+e^{-\frac{x}{2}}\right)$

$\equiv \frac{1}{2}(e^{x}-e^{-x})\equiv\frac{e^{x}-e^{-x}}{2}\equiv\sinh x$:LHS

69)

LHS: $\cosh^2\left(\frac{x}{3}\right)-4\sinh^2\left(\frac{x}{6}\right)\cosh^2\left(\frac{x}{6}\right)$

$\equiv \left(\frac{e^{\frac{x}{3}}+e^{-\frac{x}{3}}}{2}\right)^2-4\left(\frac{e^{\frac{x}{6}}-e^{-\frac{x}{6}}}{2}\right)^2\left(\frac{e^{\frac{x}{6}}+e^{-\frac{x}{6}}}{2}\right)^2$

$\equiv \frac{1}{4}\left(e^{\frac{x}{3}}+e^{-\frac{x}{3}}\right)^2-\frac{1}{4}\left(e^{\frac{x}{6}}-e^{-\frac{x}{6}}\right)^2\left(e^{\frac{x}{6}}+e^{-\frac{x}{6}}\right)^2$

$\equiv \frac{1}{4}\left(e^{\frac{x}{3}}+e^{-\frac{x}{3}}\right)^2-\frac{1}{4}\left(\left(e^{\frac{x}{6}}-e^{-\frac{x}{6}}\right)\left(e^{\frac{x}{6}}+e^{-\frac{x}{6}}\right)\right)^2$

$\equiv \frac{1}{4}\left(e^{\frac{x}{3}}+e^{-\frac{x}{3}}\right)^2-\frac{1}{4}\left(e^{\frac{x}{3}}-e^{-\frac{x}{3}}\right)^2$

$\equiv \frac{1}{4}\left(\left(e^{\frac{x}{3}}+e^{-\frac{x}{3}}\right)^2-\left(e^{\frac{x}{3}}-e^{-\frac{x}{3}}\right)^2\right)$

$$\equiv \frac{1}{4}\left(e^{\frac{2x}{3}} + 2 + e^{-\frac{2x}{3}} - e^{\frac{2x}{3}} + 2 - e^{-\frac{2x}{3}}\right)$$
$$\equiv \frac{1}{4}(4) \equiv 1 : \text{RHS}$$

70)

RHS: $4\cosh^4\left(\frac{x}{4}\right) - \sinh^2\left(\frac{x}{2}\right)\tanh^2\left(\frac{x}{4}\right)$

$$\equiv 4\left(\frac{e^{\frac{x}{4}}+e^{-\frac{x}{4}}}{2}\right)^4 - \left(\frac{e^{\frac{x}{2}}-e^{-\frac{x}{2}}}{2}\right)^2\left(\frac{e^{\frac{x}{4}}-e^{-\frac{x}{4}}}{e^{\frac{x}{4}}+e^{-\frac{x}{4}}}\right)^2$$
$$\equiv \frac{1}{4}\left(e^{\frac{x}{4}} + e^{-\frac{x}{4}}\right)^4 - \frac{1}{4}\left(e^{\frac{x}{2}} - e^{-\frac{x}{2}}\right)^2\left(\frac{e^{\frac{x}{4}}-e^{-\frac{x}{4}}}{e^{\frac{x}{4}}+e^{-\frac{x}{4}}}\right)^2$$
$$\equiv \frac{1}{4}\left(e^{x} + 4e^{\frac{x}{2}} + 6 + 4e^{-\frac{x}{2}} + e^{-x}\right) - \frac{1}{4} \times \frac{\left(e^{\frac{x}{2}}-e^{-\frac{x}{2}}\right)^2\left(e^{\frac{x}{4}}-e^{-\frac{x}{4}}\right)^2}{\left(e^{\frac{x}{4}}+e^{-\frac{x}{4}}\right)^2}$$
$$\equiv \frac{1}{4}e^{x} + e^{\frac{x}{2}} + \frac{3}{2} + e^{-\frac{x}{2}} + \frac{1}{4}e^{-x} - \frac{1}{4} \times \frac{\left(\left(e^{\frac{x}{4}}+e^{-\frac{x}{4}}\right)\left(e^{\frac{x}{4}}-e^{-\frac{x}{4}}\right)\right)^2\left(e^{\frac{x}{4}}-e^{-\frac{x}{4}}\right)^2}{\left(e^{\frac{x}{4}}+e^{-\frac{x}{4}}\right)^2}$$
$$\equiv \frac{1}{4}e^{x} + e^{\frac{x}{2}} + \frac{3}{2} + e^{-\frac{x}{2}} + \frac{1}{4}e^{-x} - \frac{1}{4} \times \frac{\left(e^{\frac{x}{4}}+e^{-\frac{x}{4}}\right)^2\left(e^{\frac{x}{4}}-e^{-\frac{x}{4}}\right)^2\left(e^{\frac{x}{4}}-e^{-\frac{x}{4}}\right)^2}{\left(e^{\frac{x}{4}}+e^{-\frac{x}{4}}\right)^2}$$
$$\equiv \frac{1}{4}e^{x} + e^{\frac{x}{2}} + \frac{3}{2} + e^{-\frac{x}{2}} + \frac{1}{4}e^{-x} - \frac{1}{4}\left(e^{\frac{x}{4}} - e^{-\frac{x}{4}}\right)^4$$
$$\equiv \frac{1}{4}e^{x} + e^{\frac{x}{2}} + \frac{3}{2} + e^{-\frac{x}{2}} + \frac{1}{4}e^{-x} - \frac{1}{4}\left(e^{x} - 4e^{\frac{x}{2}} + 6 - 4e^{-\frac{x}{2}} + e^{-x}\right)$$
$$\equiv \frac{1}{4}e^{x} + e^{\frac{x}{2}} + \frac{3}{2} + e^{-\frac{x}{2}} + \frac{1}{4}e^{-x} - \frac{1}{4}e^{x} + e^{\frac{x}{2}} - \frac{3}{2} + e^{-\frac{x}{2}} - \frac{1}{4}e^{-x}$$
$$\equiv 2e^{\frac{x}{2}} + 2e^{-\frac{x}{2}} \equiv 4\left(\frac{e^{\frac{x}{2}}+e^{-\frac{x}{2}}}{2}\right) \equiv 4\cosh\left(\frac{x}{2}\right) : \text{LHS}$$